PowerSolv

Distribution System Application Software (DSAS)

Data Description





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PowerSolv DSAS Data Description

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DSL Template for Feeder

FEEDER DATA REQUIREMENTS TEMPLATES

The data input required for the Segregation of Distribution System Losses into Technical and Non-Technical Loss and to establish the Recoverable Rate of Distribution System Losses shall be submitted to the using the following Templates:

a. DSL-00 Simulation Parameters
b. DSL-01 Customer Data
c. DSL-02 Billing Cycle
d. DSL-03 Customer Energy Consumption Data
e. DSL-04 Load Curve Data
f. DSL-05 Bus Data
g. DSL-06 Primary Distribution Line Data (Overhead)
h. DSL-07 Primary Distribution Line Data (Underground Cable)
i. DSL-08 Primary Customer Service Drop Data (Overhead)
j. DSL-09 Primary Customer Service Drop Data (Underground Cable)
k. DSL-10 Distribution Transformer Data
l. DSL-11 Secondary Distribution Line Data
m. DSL-12 Secondary Customer Service Drop Data
n. DSL-13 Voltage Regulator Data
o. DSL-14 Shunt Capacitor Data
p. DSL-15 Shunt Inductor Data
q. DSL-16 Series Inductor Data
r. DSL-17 Generator Data
s. DSL-18 Generator Dispatch Curve
t. DSL-19 Voltage Profile
u. DSL-20 Generator Reactive Power Curve

DSL-00: Simulation Parameters

Feeder Name
Specify the ID that will identify the feeder-level distribution system. Use up to 25 alphanumeric characters without spaces.

Feeder Root Bus ID
Specify the Bus ID of the root bus or root node of the distribution system.

Feeder Energy Input (kWh)
Specify the Input Energy supplied at the root bus in kWh.

DU Use (kWh)
Specify the administrative use of the distribution utility in kWh.

Power Mismatch
Specify the power mismatch that will be used as a convergence index.

Base kVA
Specify the base kVA of the system.

Maximum Iteration
Specify the number or maximum iteration that will be used in the simulation.

Percent PQ
Specify the percentage of Constant Power loads in the system. (Use 0 to 100)

Percent Z
Specify the percentage of Constant Power loads in the system. (Use 0 to 100)

Percent I
Specify the percentage of Constant Power loads in the system. (Use 0 to 100)

Percent Loading
Specify the percentage loading of the system.

DSL-01: Customer Data

Customer ID
Specify the unique ID that will identify a customer (e.g., Customer Account Number). All customers must be included in the list.

Customer Name
Specify the name of the Customer that corresponds to the Customer ID.

Customer Type
Specify the customer type or classification code using up to 25 characters (e.g., RES1 for small residential, RES2 for large residential, etc.).

Service Voltage
Specify the nominal service voltage being supplied to the customer in Volts (e.g., 230 volts).

Phase
Specify the number of phase(s) of the customer service:

1 – Single-Phase
3 – Three-Phase

DSL-02: Billing Cycle

Billing Period Code
Specify the Billing Period according to the following coding system:

YYYYMM
Where, YYYY – Year of the meter reading (billing) period (e.g., 2001 for year 2001)
MM – month of the meter reading (billing) period (e.g., 08 for August)

Period Covered
Specify the Month, day and year covered by the Billing Cycle.

Number of Days
Specify the number of days covered by the Billing Period.

Number of Hours
Specify the total number of hours covered by the Billing Period.

DSL-03: Customer Energy Consumption Data

Customer ID
Specify the unique ID that identifies a customer. This must be the same ID used in the Customer Data.

Billing Period Code
Specify the Billing Period according to the following coding system:

YYYYMM
Where, YYYY – Year of the meter reading (billing) period
(e.g., 2001 for year 2001)
MM – month of the meter reading (billing) period
(e.g., 08 for August)

Energy Consumed (kWHr)
Specify the kWHr reading of the electric meter of the customer for the billing period.

Power Factor
Specify the average power factor (measured or estimated) of the customer for the billing period.

Status
Specify the status of the customer connection (1 if connected, 0 if disconnected).

DSL-04: Load Curve Data

Load Curve ID
Specify the unique ID of the load curve for the Customer Type. It is possible to have different load curves for weekdays and weekend or for seasonal variations for each type or sub-type of customer.

Customer Type
Specify the customer classification code represented by the load curve. This must be the same Customer Type specified in the Customer Data and Meter Reading Data.

Description
Describe the Customer Type. For example, RES1 is a small residential customer with monthly consumption of not more than 50 kWhr.

Hour 1 to Hour 24
Specify the normalized hourly demand from Hour 1 to Hour 24 of the Load Curve in per unit. This can be obtained by monitoring the 24-hour demand pattern of the Customer Type (e.g., hourly Amps, kW, kVA, etc.). To obtain the Normalized demand in per unit (p.u.), each hourly demand is divided by the peak demand. Thus, the highest value of the Normalized Hourly Demand is 1.0 which coincides with the peak hour.

DSL-05: Bus Data

Bus ID
Enter the unique identification (ID) of the Bus or Node in the Distribution System using up to 25 alphanumeric characters. Bus or node is created for each connection or junction point from Sub-transmission lines to Secondary Distribution Lines (e.g., connection point between transformer terminal and a line segment, connection point between two line segments, etc.).

Bus Description
Enter the Description of the Bus or Node (e.g., Transformer TX-1 primary).

Nominal Voltage (kV)
Specify the nominal voltage of the Bus or Node in kV (e.g., 13.2, 0.48).

DSL-06: Primary Distribution Line Data – Overhead

Each Primary Distribution Line segment (i.e., the section of the Primary Distribution Line that can be identified by only one sending-end and only one receiving-end) must be included as one data entry. The whole length of the Distribution Line may be entered as one or more line segments depending on the connection points and the construction arrangement (e.g., loop, expanded radial, etc.).

Connection point is created if an equipment (e.g., Shunt Capacitor), line, or load is connected to the Distribution Line. This connection point must be assigned a Bus ID. *[Note: Unless a "Connection Point" is created, a "Pole-to-Pole" line segment should not be treated as a Primary Distribution Line Segment to avoid increasing the number of Buses]*

Primary Distribution Line Segment ID
Enter the unique ID of the Primary Distribution Line segment using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the sending end of the Primary Distribution Line segment.

To Bus ID
Specify the Bus ID of the receiving end of the Primary Distribution Line segment.

Phasing
To distinguish the Primary Distribution Lines with grounded Neutral Wire from those without grounded wire, the following Phasing convention shall be used:

a) For Uni-grounded Distribution System: ABC; and
b) For Multi-grounded Distribution System: ABCN

In specifying the Phasing, conductor 1 shall be assumed to be always in the leftmost side when facing the secondary side of the Substation Power Transformer for triangular and horizontal configuration or the highest conductor in the case of vertical configuration. Phasing shall be specified using the following conventions (see Figure B-2):

a) ABCN or ABC if Phases A, B and C correspond to conductor 1, 2 and 3, respectively;
b) ACBN or ACB if Phases A, C and B correspond to conductor 1, 2 and 3, respectively;
c) BCAN or BCA if Phases B, C and A correspond to conductor 1, 2 and 3, respectively;
d) BACN or BAC if Phases B, A and C correspond to conductor 1, 2 and 3, respectively;
e) CABN or CAB if Phases C, A and B correspond to conductor 1, 2 and 3, respectively;
f) CBAN or CBA if Phases C, B and A correspond to conductor 1, 2 and 3, respectively;
g) ABN or AB if Phases A and B correspond to conductor 1 and 2, respectively;
h) BAN or BA if Phases B and A correspond to conductor 1 and 2, respectively;
i) BCN or BC if Phases B and C correspond to conductor 1 and 2, respectively;
j) CBN or CB if Phases C and B correspond to conductor 1 and 2, respectively;

k) CAN or CA if Phases C and A correspond to conductor 1 and 2, respectively;
l) ACN or AC if Phases A and C correspond to conductor 1 and 2, respectively;
m) AN if Phase A corresponds to conductor 1;
n) BN if Phase A corresponds to conductor 1; and
o) CN if Phase A corresponds to conductor 1

Configuration

Specify installation configuration of conductors (see Figure B-1):

Triangular;
Horizontal; or
Vertical.

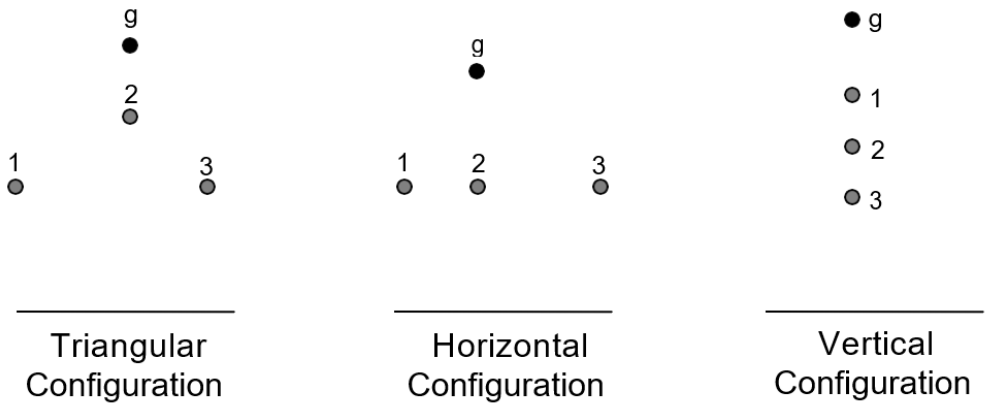


Figure B-1. Conductor Configuration

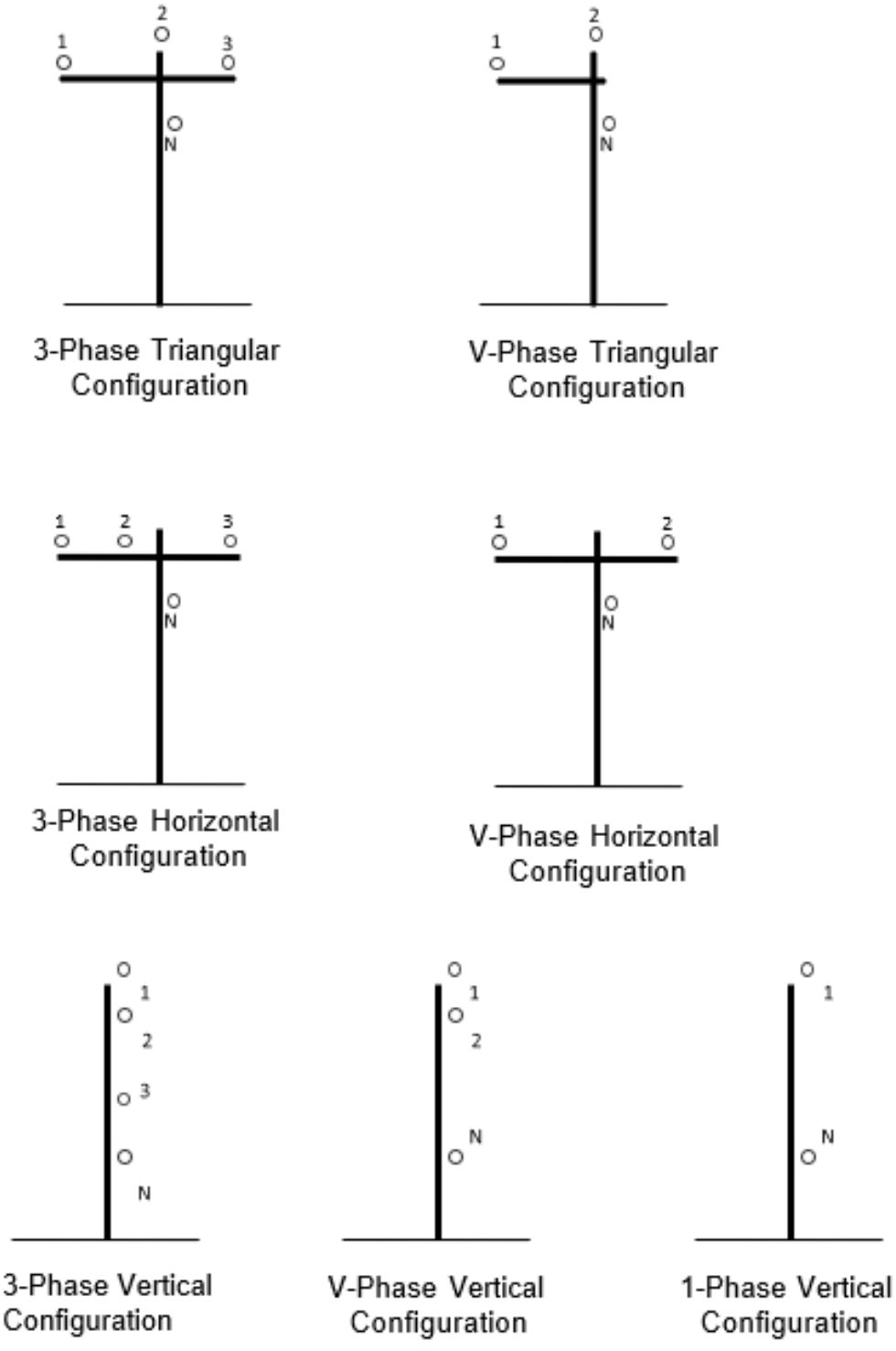


Figure B-2. Conductor Arrangement of Primary Distribution Line

System Grounding Type

Specify the system grounding type:

Uni-grounded ; or Multi-grounded.

Length (meters)

Enter the length of the Primary Distribution Line segment in meters.

Conductor Type

Specify the material type of the phase conductor:

ACSR	–	for Aluminum Cable Steel Reinforced; or
AL	–	for Aluminum Conductor
CU	–	for Copper Conductor

Conductor Size and Unit (C)

Specify size of phase conductors in AWG, CM or mm2

Strands (C)

Specify the number of strands of the phase conductors. For ACSR, the number of strands of the aluminum and steel shall be specified according to the following format:

Al/St

For example, an ACSR with 6 Aluminum strands and 1 Steel should have “6/1” entry in the Strands column.

Neutral Wire Type

Specify the material type of the Neutral Wire:

ACSR	–	for Aluminum Cable Steel Reinforced; or
AL	–	for Aluminum Conductor
CU	–	for Copper Conductor

Neutral Wire Size and Unit (NW)

Specify size of Neutral Wire in AWG, CM or mm2

Strands (NW)

Specify the number of strands of the Neutral Wire. For ACSR, the number of strands of the aluminum and steel shall be specified according to the following format:

Al/St

For example, an ACSR with 6 Aluminum strands and 1 Steel should have “6/1” entry in the Strands column.

Spacing D12 (meters)

Specify the distance in meters between Conductor 1 (leftmost conductor for triangular and horizontal configuration or the highest conductor for vertical configuration) and Conductor 2 (middle conductor). This Phasing convention shall translate to the following conductors spacing (see Figure B-3):

a) For ABCN and ABC: D12 is the distance between Phase A and Phase B;
b) For ACBN or ACB: D12 is the distance between Phase A and Phase C;
c) For BCAN or BCA: D12 is the distance between Phase B and Phase C;
d) For BACN or ABC: D12 is the distance between Phase B and Phase A;
e) For CABN or CAB: D12 is the distance between Phase C and Phase A;
f) For CBA or CBA: D12 is the distance between Phase C and Phase B;
g) For ABN or AB: D12 is the distance between Phase A and Phase B;
h) For BAN or BA: D12 is the distance between Phase B and Phase A;
i) For BCN or BC: D12 is the distance between Phase B and Phase C;
j) For CBN or CB: D12 is the distance between Phase C and Phase B;
k) For CAN or CA: D12 is the distance between Phase A and Phase B;
l) For ACN or AC: D12 is the distance between Phase A and Phase C;
m) For AN: D12 = 0;
n) For BN: D12 = 0;
o) For CN: D12 =0

Spacing D23 (meters)

Specify the distance in meters between Conductor 2 (middle conductor) and Conductor 3 (rightmost conductor for triangular and horizontal configuration or the lowest conductor for vertical configuration). This Phasing convention shall translate to the following conductors spacing (see Figure B-3):

a) For ABCN and ABC: D23 is the distance between Phase B and Phase C;
b) For ACBN or ACB: D23 is the distance between Phase C and Phase B;
c) For BCAN or BCA: D23 is the distance between Phase C and Phase A;
d) For BACN or ABC: D23 is the distance between Phase A and Phase C;
e) For CABN or CAB: D23 is the distance between Phase A and Phase B;

f) For CBA or CBA: D23 is the distance between Phase B and Phase A;
g) For ABN or AB: D23 = 0;
h) For BAN or BA: D23 = 0;
i) For BCN or BC: D23 = 0;
j) For CBN or CB: D23 = 0;
k) For CAN or CA; D23 = 0;
l) For ACN or AC; D23 = 0;
m) For AN: D23 = 0;
n) For BN: D23 = 0; and
o) For CN: D23 = 0.

Spacing D13 (meters)

Specify the distance in meters between Conductor 1 (leftmost conductor for triangular and horizontal configuration or the highest conductor for vertical configuration) and Conductor 3 (rightmost conductor for triangular and horizontal configuration or the lowest conductor for vertical configuration). This Phasing convention shall translate to the following conductors spacing (see Figure B-3):

a) For ABCN and ABC: D13 is the distance between Phase A and Phase C;
b) For ACBN or ACB: D13 is the distance between Phase A and Phase B;
c) For BCAN or BCA: D13 is the distance between Phase B and Phase A;
d) For BACN or ABC: D13 is the distance between Phase B and Phase C;
e) For CABN or CAB: D13 is the distance between Phase C and Phase B;
f) For CBA or CBA: D13 is the distance between Phase C and Phase A
g) For ABN or AB: D13 = 0;
h) For BAN or BA: D13 = 0;
i) For BCN or BC: D13 = 0;
j) For CBN or CB: D13 = 0;
k) For CAN or CA: D13 = 0;
l) For ACN or AC: D13 = 0;
m) For AN: D13 = 0;
n) For BN: D13 = 0; and
o) For CN: D13 = 0.

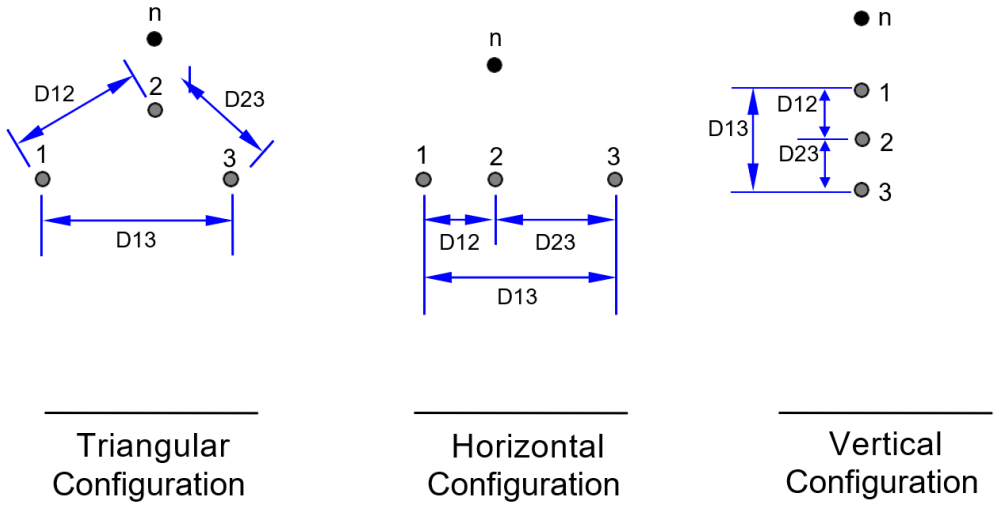


Figure B-3. Conductor Spacing

Spacing D1n (meters)
Specify the distance in meters between Conductor 1 and the Neutral Wire (see Figure B-4).

Spacing D2n (meters)
Specify the distance in meters between Conductor 2 and the Neutral Wire (see Figure B-4).

Spacing D3n (meters)
Specify the distance in meters between Conductor 3 and the Neutral Wire (see Figure B-4).

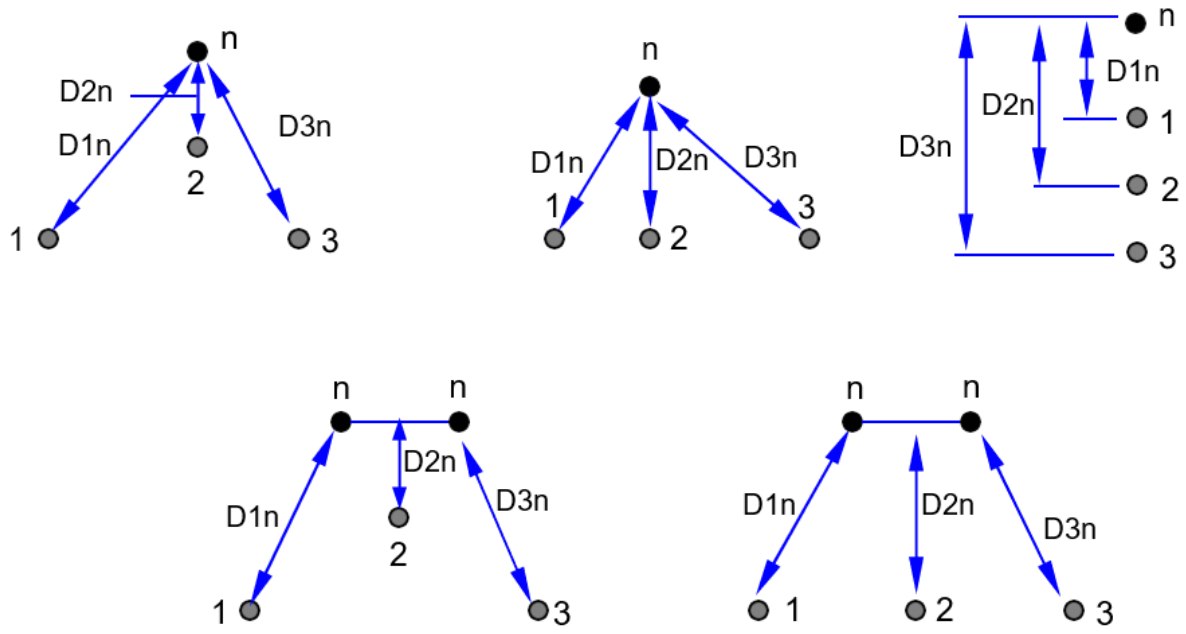


Figure B-4. Spacing of Phase Conductors and Ground Wire

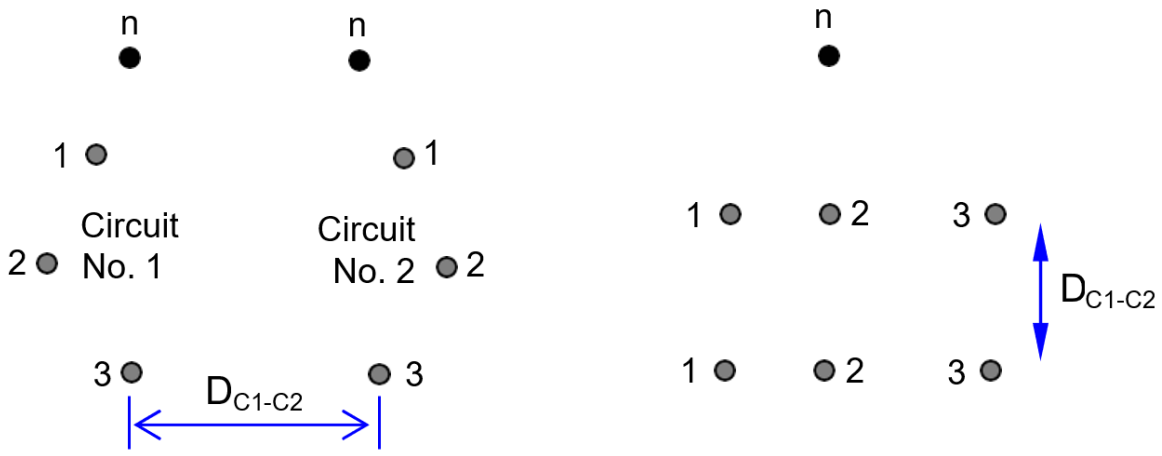


Figure B-5. Distance Between Circuit 1 and Circuit 2

Spacing DC1-C2 (meters)
For Parallel configuration (Double Circuit), specify the distance in meters between nearest phase conductors of Circuit 1 and Circuit 2 (see Figure B-5).

Height H1 (meters)
Specify the height of Conductor 1 of the Primary Distribution Line Segment from the earth in meters. Enter “0.0” if not applicable (see Figure B-6).

Height H2 (meters)
Specify the height of Conductor 2 of the Primary Distribution Line Segment from the earth in meters. Enter “0.0” if not applicable (see Figure B-6).

Height H3 (meters)
Specify the height of Conductor 3 of the Primary Distribution Line Segment from the earth in meters. Enter “0.0” if not applicable (see Figure B-6).

Height Hn (meters)
Specify the height of the Neutral Wire from the earth in meters. Enter “0.0” if not applicable (see Figure B-6).

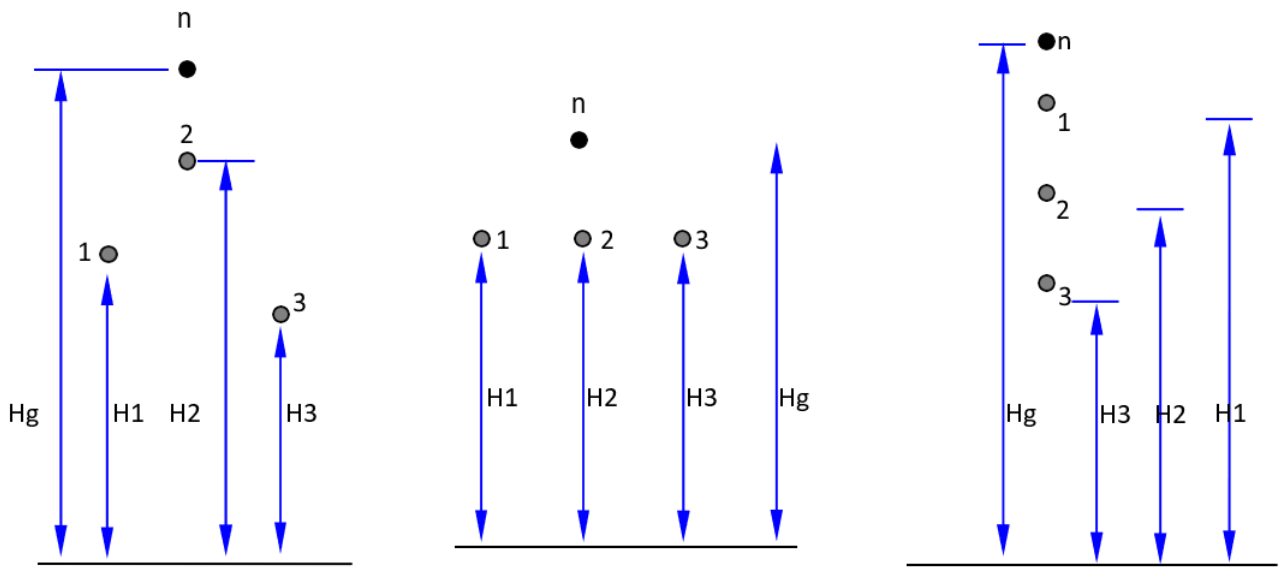


Figure B-6. Height of Phase Conductors and Ground Wire

Earth Resistivity (Ohm-meter)

Specify the earth resistivity in ohm-meter. Use 100 ohm-meters for average damp earth if the value of resistivity is not known.

DSL-07: Primary Distribution Line Data – Underground Cable

Primary Distribution Line Segment ID
Enter the unique ID of the Primary Distribution Line Line segment using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the sending end of the sub-transmission Line segment.

To Bus ID
Specify the Bus ID of the receiving end of the sub-transmission Line segment.

Phasing
Specify the phase arrangement of the Sub-transmission Line Segment:

ABC, ACB, BCA, BAC, or CAB or CBA

In specifying the Phasing, conductor 1 shall be assumed to be always in the leftmost side when facing the secondary side of the Substation Power Transformer.

Length (meters)
Enter the length of the Sub-transmission Line segment in meters. Conductor Type
Specify the material type of the phase conductor:

AL – for Aluminum Conductor; or
CU – for Copper Conductor.

Conductor Size and Unit (C)
Specify size of phase conductors in AWG, CM or mm^2

No. of Cores
Specify the number of cores of the cable:

1 – Single-Core Cable
2 – Two-Core Cable
3 – Three-Core Cable
4 – Four Core Cable Diameter under Armour (mm)

Specify the diameter under the Armour Wire in mm (see Figure B-6).

Armour Wire Diameter (mm)
Specify the diameter of the Armour Wire in mm (see Figure B-6).

Overall Diameter (mm)
Specify the overall diameter of the Cable in mm (see Figure B-6).

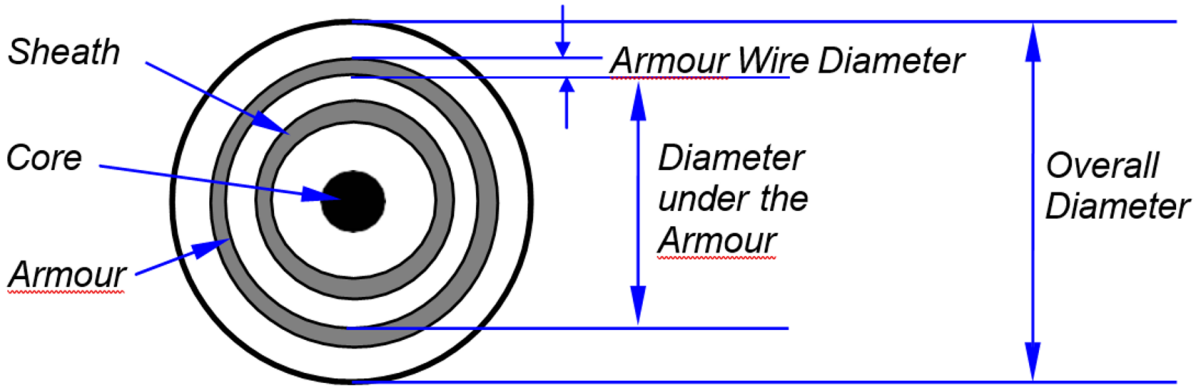


Figure B-7. Constructional Data of Underground Cable

AC Resistance (ohm/Km)
Specify the AC resistance of the conductor in ohm/Km.

Inductive Reactance (Ohm/Km)
Specify the inductive reactance of the cable in ohm/Km.

Capacitance (Micro-farad/Km)
Specify the star capacitance of the cable in Micro-farad/Km.

Earth Resistivity (ohm-meter)
Specify the earth resistivity in ohm-meter. Use 100 ohm-meters for average damp earth if the value of resistivity is not known.

DSL-08: Primary Customer Service Drop Data – Overhead

The Primary Customer Service Drop is the conductor extended from the Primary Distribution Line to the customer service entrance. The data for the Overhead Primary Distribution Customer Service Drops are the same as the requirements for the Overhead Primary Distribution Line except for the Primary Customer Service Drop ID.

DSL-09: Primary Customer Service Drop Data – Underground Cable

The data for the Underground Primary Distribution Customer Service Drops are the same as the requirements for the Underground Primary Distribution Line except for the Primary Customer Service Drop ID.

DSL-10: Distribution Transformer Data

Distribution Transformer ID
Specify the unique ID for the Distribution Transformer using up to 25 alphanumeric characters.

From Primary Bus ID
Specify the Bus ID where the primary of the Distribution Transformer is connected.

To Secondary Bus ID
Specify the Bus ID where the secondary of the Distribution Transformer is connected.

Phasing
Specify the Phasing of the Distribution Transformer (at the secondary terminals):

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for Multi-grounded Three- Phase Transformer bank;
b) ABN, BAN, BCN, CBN, CAN or ACN for Multi-grounded V-Phase Transformer bank;
c) AN, BN or CN for Multi-grounded Single-Phase Transformer
d) ABC, ACB, BCA, BAC, CAB, CBA for Uni-grounded Three-Phase Transformer bank;
e) AB, BA, BC, CB, CA or AC for Uni-grounded Single-Phase Transformer;

Installation Type

Specify the Installation Type of the Distribution Transformer:

Pole-mounted or Pad-mounted

No. of DTs in Bank

Specify the number of Distribution Transformer in bank:

1 – One (1) Single-phase Distribution Transformer
2 – Two (2) single-phase Distribution Transformers
3 – One (1) Three-phase Distribution Transformer
4 – Three (3) single-phase Distribution Transformers

Connection

Specify the connection of the Distribution Transformer:

1 – Single Phase
2 – Delta–Delta
3 – Delta–Wye Ground
4 – Delta–Wye
5 – Wye Ground–Wye Ground
6 – Wye Ground–Wye
7 – Wye–Wye Ground
8 – Wye–Wye
9 – Wye Ground–Delta
10 – Wye–Delta
11 – Open Delta–Open Delta
12 – Open Wye–Open Delta

kVA Rating

Specify the Rated Capacity of the Distribution Transformer in kVA. For two (2) or three (3) single-phase transformers in a bank, the rated KVA of the largest Distribution Transformer shall be used.

Primary Voltage Rating (kV)

Specify the Voltage Rating of the primary winding of the Distribution Transformer in kV. The voltage rating should be taken from the nameplate and not the resulting line-to-line voltage of the transformer bank.

Secondary Voltage Rating (kV)
Specify the Voltage Rating of the secondary winding of the Distribution Transformer in kV. The voltage rating should be taken from the nameplate and not the resulting line-to-line voltage of the transformer bank.

Primary Tap Voltage (kV)
Specify the Primary Tap Voltage of the Distribution Transformer in kV. Enter the Rated Primary Voltage if the Distribution Transformer has no taps in the primary.

Secondary Tap Voltage (kV)
Specify the Secondary Tap Voltage of the Distribution Transformer in kV. Enter the Rated Secondary Voltage if the Distribution Transformer has no taps in the secondary.

%Z
Specify the percent impedance (%Z) of the Distribution Transformer taken from the nameplate. Use typical value if %Z is not available.

X/R Ratio
Specify the X/R Ratio of the impedance of the Distribution Transformer. Use typical value if X/R Ratio is not available.

No-Load Loss (kW)
Specify the No-load loss of the Distribution Transformer in kW. Use typical value if X/R Ratio is not available.

Exciting Current (%)
Specify the exciting current of the Distribution Transformer in percent (%) of the rated current. Use typical value if exciting current is not available.

DSL-11: Secondary Distribution Line Data

Each Secondary Distribution Line segment (i.e., the section of the Secondary Distribution Line that can be identified by only one sending-end and only one receiving-end) must be included as one data entry. The whole length of the Secondary Line may be entered as one or more line segments depending on the connection points created by secondary lateral lines or service drops.

Secondary Distribution Line ID

Enter the unique ID of the Secondary Distribution Line segment using up to 25 alphanumeric characters.

From Bus ID

Specify the Bus ID of the sending end of the Secondary Distribution Line segment.

To Bus ID

Specify the Bus ID of the receiving end of the Secondary Distribution Line segment.

Phasing

Specify the Phasing of the Secondary Distribution Line:

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for 4-wire three-phase Secondary System;
b) ABC, ACB, BCA, BAC, CAB, CBA for 3-wire three-Phase Secondary System;
c) ABN, BAN, BCN, CBN, CAN or ACN for 3-wire single-phase Secondary System;
d) AN, BN or CN for 2-wire single-phase Secondary System;

Installation Type

Specify the type of installation of the secondary distribution line segment:

1 – Overhead, underbuilt
2 – Overhead, open secondary
3 – Underground in magnetic raceway (e.g., Rigid Steel Conduit)
4 – Underground in non-magnetic raceway (e.g., PVC)

Length (meters)

Specify the length of the Secondary Distribution Line segment in meters.

Conductor Type

Specify the material type of the conductors:

ACSR – for Aluminum Cable Steel Reinforced;
AL – for Aluminum Conductor; or
CU – for Copper Conductor.

Conductor Size and Unit

Specify size of phase conductors in AWG, CM or mm2

Strand
Specify the number of strands of the phase conductors. For ACSR, the number of strands of the aluminum and steel shall be specified according to the following format:

Al/St

For example, an ACSR with 6 Aluminum strands and 1 Steel should have “6/1” entry in the Strands column.

DSL-12: Secondary Customer Service Drop Data

The Secondary Customer Service Drop is the conductor extended from the Secondary Distribution Line or directly from the Distribution Transformer to the customer service entrance.

Secondary Customer Service Drop ID
Enter the unique ID of the Secondary Customer Service Drop using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the sending end of the Secondary Customer Service Drop.

To Customer ID
Specify the Customer ID of the customer that is connected at the end of the Secondary Customer Service Drop.

Phasing
Specify the Phasing of the Secondary Customer Service Drop:

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for 4-wire three-phase Secondary Service;
b) ABC, ACB, BCA, BAC, CAB, CBA for 3-wire three-Phase Secondary Service;
c) ABN, BAN, BCN, CBN, CAN or ACN for 3-wire single-phase Secondary Service; and
d) AN, BN or CN for 2-wire single-phase Secondary Service.

Installation Type
Enter the type of installation of the Secondary Customer Service Drop:

1 – Overhead (or Arial)

2 – Underground in magnetic Raceway (e.g., Rigid Steel Conduit)
3 – Underground in non-magnetic Raceway (e.g., PVC)

Length-1 (meters)
Enter the length in meters of the Secondary Customer Service Drop from the Secondary Distribution Line or form the Distribution Transformer Connection Point to the Metering Point.

Length-2 (meters)
Enter the length in meters of the Secondary Customer Service Drop from the Metering Point to Connection Point of the Customer.

Conductor Type
Specify the material type of the conductors:

ACSR	–	for Aluminum Cable Steel Reinforced;
AL	–	for Aluminum Conductor; or
CU	–	for Copper Conductor.

Conductor Size and Unit
Specify size of phase conductors in AWG, CM or mm2

Strand
Specify the number of strands of the phase conductors. For ACSR, the number of strands of the aluminum and steel shall be specified according to the following format:

Al/St

For example, an ACSR with 6 Aluminum strands and 1 Steel should have “6/1” entry in the Strands column.

DSL-13: Voltage Regulator Data

Voltage Regulator ID
Specify the unique ID for the Voltage Regulator using up to 25 alphanumeric characters.

Bus ID
Specify the Bus ID of the source side of the Voltage Regulator.

To Bus ID
Specify the Bus ID of the load side of the Voltage Regulator.

Regulated Bus ID
Specify the Bus ID of the regulating point (Bus or Node) whose voltage is being controlled by the Voltage Regulator.

Phase Type
Specify the type of Voltage Regulator:

1 – Single phase
2 – Two single phase
3 – Three-phase, gang operated
4 – Three single phase, independently operated

Phasing
Specify the Phasing of the Voltage Regulator:

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for Multi-grounded Three- Phase System;
b) ABN, BAN, BCN, CBN, CAN or ACN for Multi-grounded V-Phase System;
c) AN, BN or CN for Multi-grounded Single-Phase System;
d) ABC, ACB, BCA, BAC, CAB, CBA for Uni-grounded Three-Phase System;
e) AB, BA, BC, CB, CA or AC for Uni-grounded Single-Phase System;

Phase Sense
Specify the phase where the Voltage Sensor (PT) is installed:

A – if Phase A
B – if Phase B
C – if Phase C

kVA Rating
Specify the Rated Capacity of the Voltage Regulator in kVA.

kV Rating
Specify the voltage Rating of the Voltage Regulator in kV.

Voltage (120V base)
Specify the desired voltage (on 120-volt base) to be held by the Voltage Regulator at the regulating point (e.g., 124 volts).

Bandwidth (120V base)
Specify the voltage level tolerance of the Voltage Regulator on 120-volt base (e.g., 2.0 volts):

R-Setting Phase A
Specify the Compensator R-dial setting (i.e., the equivalent resistance to the regulating point calibrated in volts) of the Voltage Regulator in Phase A. Enter “0.0” if not applicable.

R-Setting Phase B
Specify the Compensator R-dial setting (i.e., the equivalent resistance to the regulating point calibrated in volts) of the Voltage Regulator in Phase B. Enter “0.0” if not applicable.

R-Setting Phase C
Specify the Compensator R-dial setting (i.e., the equivalent resistance to the regulating point calibrated in volts) of the Voltage Regulator in Phase C. Enter “0.0” if not applicable.

X-Setting Phase A
Specify the Compensator X-dial setting (i.e., the equivalent reactance to the regulating point calibrated in volts) of the Voltage Regulator in Phase A. Enter “0.0” if not applicable.

X-Setting Phase B
Specify the Compensator X-dial setting (i.e., the equivalent reactance to the regulating point calibrated in volts) of the Voltage Regulator in Phase B. Enter “0.0” if not applicable.

X-Setting Phase C
Specify the Compensator X-dial setting (i.e., the equivalent reactance to the regulating point calibrated in volts) of the Voltage Regulator in Phase C. Enter “0.0” if not applicable.

Primary Current Rating (A)
Specify the primary current rating of the Current Transformer used for the Voltage Regulator. The CT secondary current is assumed 1 Ampere.

PT Ratio
Specify the voltage ratio of the Potential Transformer used for the Voltage Regulator. Usually the PT secondary voltage of Voltage Regulator is 120 volts. For example, a PT rated 13,200/120 volts has a PT Ratio of 110.

No-Load Loss (kW)
Specify the No-Load (fixed) loss of the Voltage Regulator in kW.

Exciting Current (%)
Specify the exciting current of the Voltage Regulator in percent (%) of the rated current.

DSL-14: Shunt Capacitor Data

Shunt Capacitor ID
Specify the unique ID for the Shunt Capacitor using up to 25 alphanumeric characters.

Bus Connected (Bus ID)
Specify the Bus ID of the Bus or Node where the Shunt Capacitor is connected.

Phase Type
Specify the construction type of Shunt Capacitor:

1 – Single-phase Shunt Capacitor
2 – Two (2) single-phase Shunt Capacitors
3 – Three-phase Shunt Capacitor
4 – Three (3) single-phase Shunt Capacitors

Phasing
Specify the Phasing of the Shunt Capacitor:

a) ABC for Three-Phase
b) AB, BC or CA for V-Phase
c) A, B, C or Single-Phase

Voltage Rating (kV)
Specify the Voltage Rating of Shunt Capacitor in kV.

KVAR Rating Phase A
Specify reactive Power Rating in kVArs of the Shunt Capacitor in phase A.

KVAR Rating Phase B
Specify reactive Power Rating in kVArs of the Shunt Capacitor in phase B.

KVAR Rating Phase C
Specify reactive Power Rating in kVArs of the Shunt Capacitor in phase C.

Power Loss (Watts)
Specify the Power Losses of the Shunt Capacitor per phase in Watts. Use Typical Value if Power Loss data of the Shunt Capacitor is not known.

DSL-15: Shunt Inductor Data

Shunt Inductor ID
Specify the unique ID for the Shunt Inductor using up to 25 alphanumeric characters.

Bus Connected (Bus ID)
Specify the Bus ID of the Bus or Node where the Shunt Inductor is connected. Phase Type
Specify the construction type of Shunt Inductor:

1 – Single-phase Shunt Inductor
2 – Two (2) single-phase Shunt Inductors
3 – Three-phase Shunt Inductor
4 – Three (3) single-phase Shunt Inductors

Phasing
Specify the Phasing of the Shunt Inductors:

a) ABC for Three-Phase
b) AB, BC or CA for V-Phase
c) A, B, C or Single-Phase

Voltage Rating (kV)
Specify the Voltage Rating of the Shunt Inductor in kV.

Resistance Phase A (Ohms)
Specify the Resistance in Ohms of the Shunt Inductor in Phase A. Use typical values if the resistance of the Shunt Inductor is not available.

Resistance Phase B (Ohms)
Specify the Resistance in Ohms of the Shunt Inductor in Phase B. Use typical values if the resistance of the Shunt Inductor is not available.

Resistance Phase C (Ohms)
Specify the Resistance in Ohms of the Shunt Inductor in Phase C. Use typical values if the resistance of the Shunt Inductor is not available.

Reactance Phase A (Ohms)
Specify the Reactance in Ohms of the Shunt Inductor in Phase A.

Reactance Phase B (Ohms)
Specify the Reactance in Ohms of the Shunt Inductor in Phase B.

Reactance Phase C (Ohms)
Specify the Reactance in Ohms of the Shunt Inductor in Phase C.

DSL-16: Series Inductor Data

Series Inductor ID
Specify the unique ID for the Series Inductor using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the source side of the Series Inductor.

To Bus ID
Specify the Bus ID of the load side of the Series Inductor.

Phase Type
Specify the construction type of Series Inductor:

1 – One (1) Single-phase Series Inductor
2 – Two (2) single-phase Series Inductors
3 – One (1) Three-phase Series Inductor
4 – Three (3) single-phase Series Inductors

Phasing
Specify the Phasing of the Series Inductors:

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for Multi-grounded Three- Phase System;
b) ABN, BAN, BCN, CBN, CAN or ACN for Multi-grounded V-Phase System;
c) AN, BN or CN for Multi-grounded Single-Phase System;

d) ABC, ACB, BCA, BAC, CAB, CBA for Uni-grounded Three-Phase System;

e) AB, BA, BC, CB, CA or AC for Uni-grounded Single-Phase System;

Voltage Rating (kV)
Specify the Voltage Rating of the Series Inductor in kV.

Resistance Phase A (Ohms)
Specify the Resistance in Ohms of the Series Inductor in Phase A. Use typical value if the resistance of the resistance of the Series Inductor is not available.

Resistance Phase B (Ohms)
Specify the Resistance in Ohms of the Series Inductor in Phase B. Use typical value if the resistance of the resistance of the Series Inductor is not available.

Resistance Phase C (Ohms)
Specify the Resistance in Ohms of the Series Inductor in Phase C. Use typical value if the resistance of the resistance of the Series Inductor is not available.

Reactance Phase A (Ohms)
Specify the Reactance in Ohms of the Series Inductor in Phase A.

Reactance Phase B (Ohms)
Specify the Reactance in Ohms of the Series Inductor in Phase B.

Reactance Phase C (Ohms)
Specify the Reactance in Ohms of the Series Inductor in Phase C.

DSL-17: Generator Data

Generator ID
Specify the unique ID of the generator in the distribution. Use up to 25 alphanumeric characters.

Bus Connected (Bus ID)
Specify the Bus ID where the distributed generator is connected.

KVA Rating

Specify the rated capacity of the distributed generator in kVA.

KW Rating
Specify the rated real power of the generator in kW.

Minimum KVAR
Specify the minimum reactive power limit that the generator can inject in kVArs. Specify a negative sign if the generator can absorb reactive power.

Maximum KVAR
Specify the maximum reactive power limit that the generator can inject in kVArs.

Voltage Rating (kV)
Specify the voltage rating of the primary winding of the distributed generator in kV. The voltage rating should be taken from the nameplate and not the resulting line-to-line voltage of the generator.

Dispatch Curve ID
Specify the Dispatch Curve ID that will be applied to the generator. All Dispatch Curve ID must be defined in the Dispatch Curve Data. Use up to 25 alphanumeric characters.

DG Model
Specify the distributed generator model used in the simulation. Both models have constant real power (P) that will follow the corresponding dispatch curve. The defined models are:

PQ – Constant real and reactive power (P, Q). Reactive Power values follow its corresponding Reactive Power Curve

PV – Constant real power and voltage (P, V). Source voltage values follow its corresponding Voltage Profile

Voltage Profile ID
Specify the Voltage Profile ID that will be applied to the generator for DGs modeled as PV. All Voltage Profile ID must be defined in the Voltage Profile Data. Put “NA” if the DG is modeled as PQ. Use up to 25 alphanumeric characters.

Reactive Power Curve ID

Specify the Reactive Power Curve ID that will be applied to the generator for DGs modeled as PQ. All Reactive Power Curve ID must be defined in the Reactive Power Curve Data. Put “NA” if the DG is modeled as PV. Use up to 25 alphanumeric characters.

Phasing

Specify the Phasing of the Generator:

ABC, ACB, BCA, BAC, CAB, or CBA.

Connection

Specify the connection of the Generator:

1 – Wye Ground
2 – Wye
3 – Delta

Positive Sequence Resistance (Ohms)

Specify the Positive Sequence Resistance in Ohms of the Generator. Use a typical value if the resistance of the Generator is not available.

Positive Sequence Reactance (Ohms)

Specify the Positive Sequence Reactance in Ohms of the Generator. Use a typical value if the reactance of the Generator is not available.

Negative Sequence Resistance (Ohms)

Specify the Negative Sequence Resistance in Ohms of the Generator. Use a typical value if the resistance of the Generator is not available.

Negative Sequence Reactance (Ohms)

Specify the Negative Sequence Reactance in Ohms of the Generator. Use a typical value if the reactance of the Generator is not available.

Zero Sequence Resistance (Ohms)

Specify the Zero Sequence Resistance in Ohms of the Generator. Use a typical value if the resistance of the Generator is not available.

Zero Sequence Reactance (Ohms)

Specify the Zero Sequence Reactance in Ohms of the Generator. Use a typical value if the reactance of the Generator is not available.

DSL-18: Generator Dispatch Curve

Dispatch Curve ID
Specify the unique ID of the generator dispatch curve in the distribution. This must be corresponding to the Dispatch Curve ID specified in the Generator Data. Use up to 25 alphanumeric characters.

Generator Type
Specify the generator type for regulation purposes. (e.g., Solar, Wind, etc.).

Description
Describe the Dispatch Curve. For example, Day-time is a dispatch schedule for the Solar PV generator types.

Hour 1 to Hour 24
Specify the normalized rated real power percentage from Hour 1 to Hour 24 of the Dispatch Curve in decimals.

DSL-19: Voltage Profile

Voltage Profile ID
Specify the ID of the voltage profile of the root source and generators modeled as PV in the distribution. This must be corresponding to the Voltage Profile ID specified in the Generator Data. The first entry should be for the 24-hour Source Voltage. Use up to 25 alphanumeric characters.

Description
Describe the Voltage Profile. For example, Source_Voltage is a constant voltage profile of 1.0 per unit.

Hour 1 to Hour 24
Specify the normalized hourly voltage values from Hour 1 to Hour 24 of the Voltage Profile in per unit.

DSL-20: Generator Reactive Power Curve

Reactive Power Curve ID
Specify the ID of the reactive power curve of the generator modeled as PQ in the distribution. This must be corresponding to the Reactive Power Curve ID specified in the Generator Data. Use up to 25 alphanumeric characters.

Description
Describe the Reactive Power Curve. For example, Unity_pf is a constant value of 0 kVAr for a unity power factor generator.

Hour 1 to Hour 24
Specify the hourly reactive power injection of the generator from Hour 1 to Hour 24 of the Reactive Power Curve in kVArs.

DSL Template for Subtransmission

SUBSTATION AND SUBTRANSMISSION DATA REQUIREMENTS TEMPLATES

The data input required for the Segregation of Distribution System Losses into Technical and Non-Technical Loss and to establish the Recoverable Rate of Distribution System Losses shall be submitted to the using the following Templates:

a. DSLSUBT-00 DSL-SUBT Simulation Parameters
b. DSLSUBT-01 Billing Cycle Data
c. DSLSUBT-02 Metered Energy Input
d. DSLSUBT-03 Load Data
e. DSLSUBT-04 Load Energy Consumption Data
f. DSLSUBT-05 Load Curve Data
g. DSLSUBT-06 Bus Data
h. DSLSUBT-07 Subtransmission Line Data (Overhead)
i. DSLSUBT-08 Subtransmission Line Data (Underground)
j. DSLSUBT-09 Power Transformer Data (Two Winding Type)
k. DSLSUBT-10 Power Transformer Data (Three Winding Type)
l. DSLSUBT-11 Subtrans Svc Drop Data (Overhead)
m. DSLSUBT-12 Subtrans Svc Drop Data (Underground)
n. DSLSUBT-13 Voltage Regulator Data
o. DSLSUBT-14 Shunt Capacitor Data
p. DSLSUBT-15 Shunt Inductor Data
q. DSLSUBT-16 Series Inductor Data
r. DSLSUBT-17 Generator Data
s. DSLSUBT-18 Generator Dispatch Curve Data
t. DSLSUBT-19 Voltage Profile Data
u. DSLSUBT-20 Generator Reactive Power Curve Data

DSLSUBT-00: Simulation Parameters

Subtransmission Name
Specify the ID that will identify the feeder-level distribution system. Use up to 25 alphanumeric characters without spaces.

Subtransmission Root Bus ID
Specify the Bus ID of the root bus or root node of the distribution system.

Subtransmission Energy Input (kWh)
Specify the Input Energy supplied at the root bus in kWh.

DU Use (kWh)
Specify the administrative use of the distribution utility in kWh.

Power Mismatch
Specify the power mismatch that will be used as convergence index in the simulation.

Base kVA
Specify the base kVA of the system.

Maximum Iteration
Specify the number or maximum iteration that will be used in the simulation.

Percent PQ
Specify the percentage of Constant Power loads in the system. (Use 0 to 100)

Percent Z
Specify the percentage of Constant Power loads in the system. (Use 0 to 100)

Percent I
Specify the percentage of Constant Power loads in the system. (Use 0 to 100)

Percent Loading
Specify the percentage loading of the system.

DSLSUBT-01: Billing Cycle Data

Billing Period Code
Specify the Billing Period according to the following coding system:

YYYYMM

Where, YYYY – Year of the meter reading (billing) period
(e.g., 2001 for year 2001)

MM – month of the meter reading (billing) period
(e.g., 08 for August)

Period Covered
Specify the Month, day and year covered by the Billing Cycle.

Number of Days
Specify the number of days covered by the Billing Period.

Number of Hours
Specify the total number of hours covered by the Billing Period.

DSLSUBT-02: Metered Energy Input

Each metered energy input describes the entry point from the Transmission network to the Subtransmission network other than the designated Root Bus, if any.

Meter ID
Enter the unique ID of the Meter segment using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the sending end of the Meter segment.

To Bus ID
Specify the Bus ID of the receiving end of the Meter segment.

Metering Point Description
Describe the metering point or location (e.g., 34.5 kV Input).

Metered Input (kWHr)
Specify the kWHr reading of the electric meter for the billing period.

DSLSUBT-03: Load Data

Load ID
Specify the unique ID that will identify a load (e.g., Customer Account Number). All loads must be included in the list.

Load Name

Specify the name of the Load that corresponds to the Load ID.

Load Type
Specify the load type or classification code using up to 25 characters (e.g., RES1 for small residential, RES2 for large residential, etc.).

Service Voltage
Specify the nominal service voltage being supplied to the load in Volts (e.g., 230 volts).

Phase
Specify the number of phase(s) of the load service:

1 – Single-Phase
3 – Three-Phase

DSLSUBT-04: Load Energy Consumption Data

Load ID
Specify the unique ID that identifies a load. This must be the same ID used in the Load Data.

Billing Period Code
Specify the Billing Period according to the following coding system:

YYYYMM
Where, YYYY – Year of the meter reading (billing) period
(e.g., 2001 for year 2001)
MM – month of the meter reading (billing) period
(e.g., 08 for August)

Energy Consumed (kWHr)
Specify the kWHr reading of the electric meter of the load for the billing period.

Power Factor
Specify the average power factor (measured or estimated) of the load for the billing period.

Status
Specify the status of the load connection (1 if connected, 0 if disconnected).

DSLSUBT-05: Load Curve Data

Load Curve ID
Specify the unique ID of the load curve for the Load Type. It is possible to have different load curves for weekdays and weekends or for seasonal variations for each type or subtype of customer.

Customer Type
Specify the customer classification code represented by the load curve. This must be the same Load Type specified in the Load Data and Meter Reading Data.

Description
Describe the Load Type. For example, IND1 is an industrial load with monthly consumption of 1000 kWhr.

Hour 1 to Hour 24
Specify the normalized hourly demand from Hour 1 to Hour 24 of the Load Curve in per unit. This can be obtained by monitoring the 24-hour demand pattern of the Load Type (e.g., hourly Amps, kW, kVA, etc.). To obtain the Normalized demand in per unit (p.u.), each hourly demand is divided by the peak demand. Thus, the highest value of the Normalized Hourly Demand is 1.0 which coincides with the peak hour.

DSLSUBT-06: Bus Data

Bus ID
Enter the unique identification (ID) of the Bus or Node in the Distribution System using up to 25 alphanumeric characters. Bus or node is created for each connection or junction point from Sub-transmission lines to Secondary Distribution Lines (e.g., connection point between transformer terminal and a line segment, connection point between two line segments, etc.).

Bus Description
Enter the Description of the Bus or Node (e.g., Transformer TX-1 primary).

Nominal Voltage (kV)
Specify the nominal voltage of the Bus or Node in kV (e.g., 13.2, 0.48).

DSLSUBT-07: Subtransmission Line Data – Overhead

Each Subtransmission Line segment (i.e., the section of the Subtransmission Line that can be identified by only one sending-end and only one receiving-end) must be included as one data entry. The whole length of the Sub-transmission Line may be entered as one or more line segments depending on the connection points and the construction arrangement (e.g., loop, expanded radial, etc.).

Subtransmission Line Segment ID
Enter the unique ID of the Sub-transmission Line segment using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the sending end of the sub-transmission Line segment.

To Bus ID
Specify the Bus ID of the receiving end of the sub-transmission Line segment.

Phasing
Specify the phase arrangement of the Sub-transmission Line Segment (see Figure B-1):
ABC, ACB, BCA, BAC, or CAB or CBA

In specifying the Phasing, conductor 1 shall be assumed to be always in the leftmost side when facing the secondary side of the Substation Power Transformer for triangular and horizontal configuration or the highest conductor in the case of vertical configuration as illustrated in Figure B-1.

Phasing shall be specified using the following conventions:

a) ABC if Phases A, B and C correspond to conductor 1, 2 and 3, respectively;
b) ACB if Phases A, C and B correspond to conductor 1, 2 and 3, respectively;
c) BCA if Phases B, C and A correspond to conductor 1, 2 and 3, respectively;
d) BAC if Phases B, A and C correspond to conductor 1, 2 and 3, respectively;
e) CAB if Phases C, A and B correspond to conductor 1, 2 and 3, respectively; and
f) CBA if Phases C, B and A correspond to conductor 1, 2 and 3, respectively.

Configuration

Specify installation configuration of conductors (see Figure B-1):

Triangular;
Horizontal;
Vertical; or
Parallel (for Double Circuit).

No. of Ground Wires

Specify the number of ground wires:

1– for one ground wire
2 – for two ground wires

Length (meters)

Enter the length of the Subtransmission Line segment in meters.

Conductor Type

Specify the material type of the phase conductor:

ACSR – for Aluminum Cable Steel Reinforced; or
AL – for Aluminum Conductor
CU – for Copper Conductor

Conductor Size and Unit (C)

Specify size of phase conductors in AWG, CM or mm2

Strands (C)

Specify the number of strands of the phase conductors. For ACSR, the number of strands of the aluminum and steel shall be specified according to the following format:

Al/St

For example, an ACSR with 6 Aluminum strands and 1 Steel should have “6/1” entry in the Strands column.

Bundled Conductors

Specify the number of bundled conductors of the phase conductors (see Figure B-7):

1 – for Single Conductor
2 – for Two-Conductor Bundle
3 – for Three-Conductor Bundle

4 – for Four-Conductor Bundle

Bundled Cond. Spacing (cm)

Specify the spacing S of bundled conductors in cm (see Figure B-7). Enter “0.0” for Single Conductor.

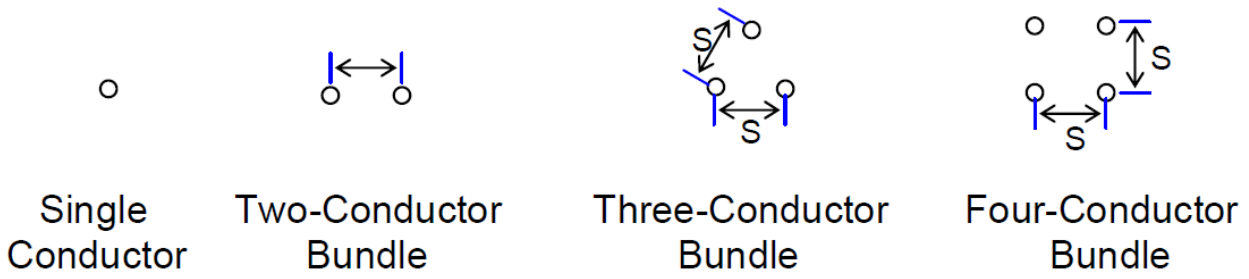


Figure B-7. Bundling of Conductors

Ground Wire Type

Specify the material type of the Ground Wire:

ACSR – for Aluminum Cable Steel Reinforced;
AL – for Aluminum Conductor;
CU – for Copper Conductor; or
ST – for Steel Wire

Ground Wire Size and Unit (GW)

Specify size of Ground Wire in AWG, CM or mm2

Strands (GW)

Specify the number of strands of the Ground Wire. For ACSR, the number of strands of the aluminum and steel shall be specified according to the following format:

Al/St

For example, an ACSR with 6 Aluminum strands and 1 Steel should have “6/1” entry in the Strands column.

Spacing D12 (meters)

Specify the distance in meters between Conductor 1 (leftmost conductor for triangular and horizontal configuration or the highest conductor for vertical configuration) and Conductor 2 (middle conductor) as illustrated in Figure B-8.

Spacing D23 (meters)
Specify the distance in meters between Conductor 2 (middle conductor) and Conductor 3 (rightmost conductor for triangular and horizontal configuration or the lowest conductor for vertical configuration) as illustrated in Figure B-8.

Spacing D13 (meters)
Specify the distance in meters between Conductor 1 (leftmost conductor for triangular and horizontal configuration or the highest conductor for vertical configuration) and Conductor 3 (rightmost conductor for triangular and horizontal configuration or the lowest conductor for vertical configuration) as illustrated in Figure B-8.

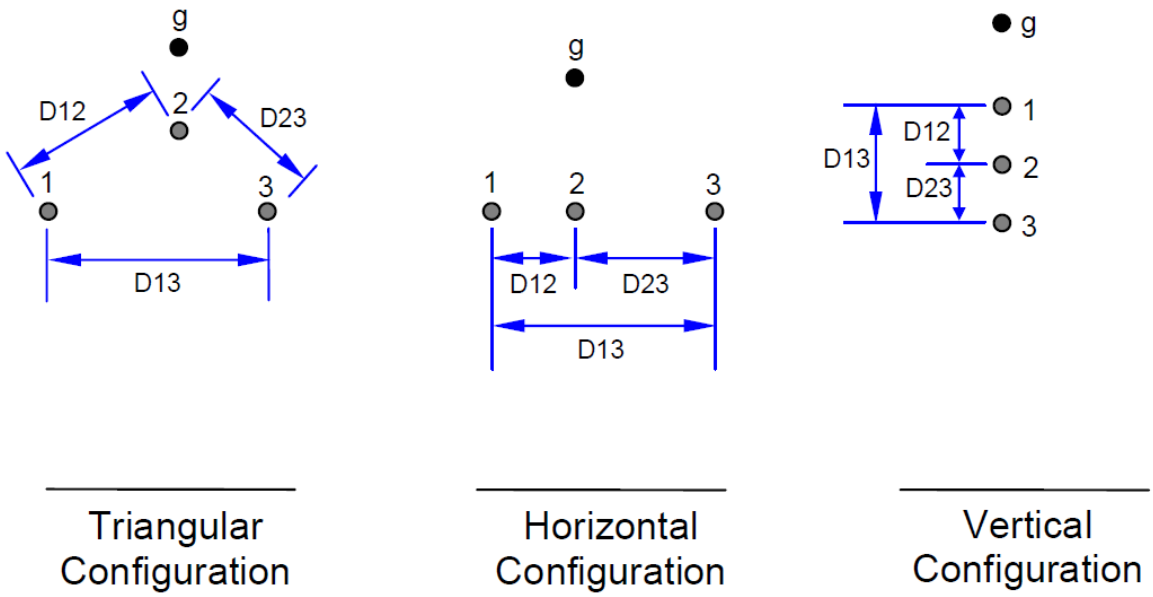


Figure B-8. Conductor Spacing

Spacing D1g (meters)
Specify the distance in meters between Conductor 1 and the Ground Wire. For Subtransmission Line with two Ground Wires, the distance of Conductor 1 to the leftmost GroundWire shall be specified (see Figure B-9).

Spacing D2g (meters)

Specify the distance in meters between Conductor 2 and the Ground Wire. For Subtransmission Line with two Ground Wires, the distance of Conductor 2 to the center of the two Ground Wires shall be specified (see Figure B-9).

Spacing D3g (meters)
Specify the distance in meters between Conductor 3 and the Ground Wire. For Subtransmission Line with two Ground Wires, the distance of Conductor 3 to the rightmost Ground Wire shall be specified (see Figure B-9).

Spacing Dgg (meters)
Specify the distance in meters between the two Ground Wires (see Figure B-10).

Spacing DC1-C2 (meters)
For Parallel configuration (Double Circuit), specify the distance in meters between nearest phase conductors of Circuit 1 and Circuit 2 (see Figure B-11).

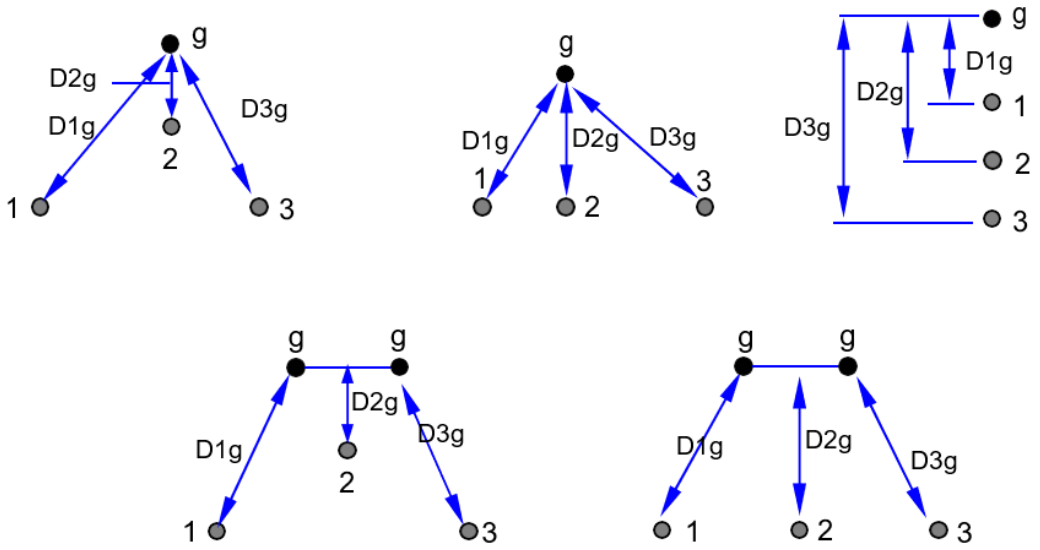


Figure B-9. Spacing of Phase Conductors and Ground Wire

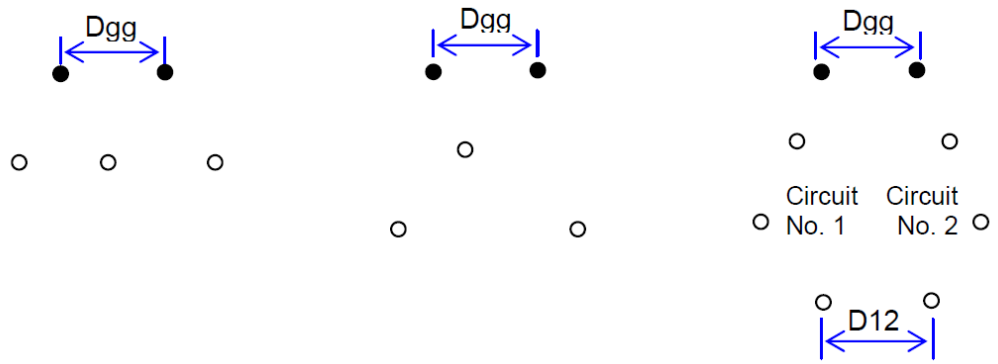


Figure B-10. Distance Between Ground Wires

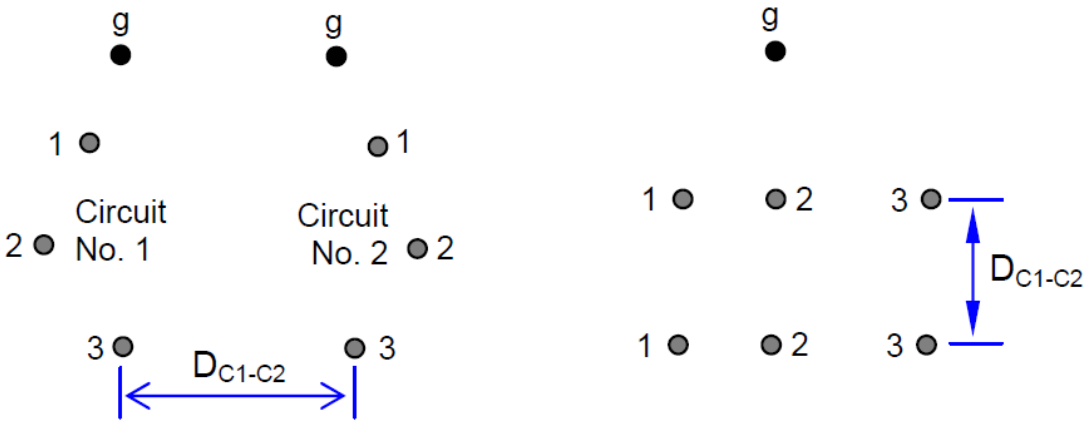


Figure B-11. Distance Between Circuit 1 and Circuit 2

Height H1 (meters)
Specify the height of Conductor 1 of the Sub-transmission Line Segment from the earth in meters (see Figure B-12). Enter “0.0” if not applicable.

Height H2 (meters)
Specify the height of Conductor 2 of the Sub-transmission Line Segment from the earth in meters (see Figure B-12). Enter “0.0” if not applicable.

Height H3 (meters)
Specify the height of Conductor 3 of the Sub-transmission Line Segment from the earth in meters (see Figure B-12). Enter “0.0” if not applicable.

Height Hg (meters)

Specify the height of the Ground Wire from the earth in meters (see Figure B-7). Enter “0.0” if not applicable.

Earth Resistivity (Ohm-meter)
Specify the earth resistivity in ohm-meter. Use 100 ohm-meters for average damp earth if the value of resistivity is not known.

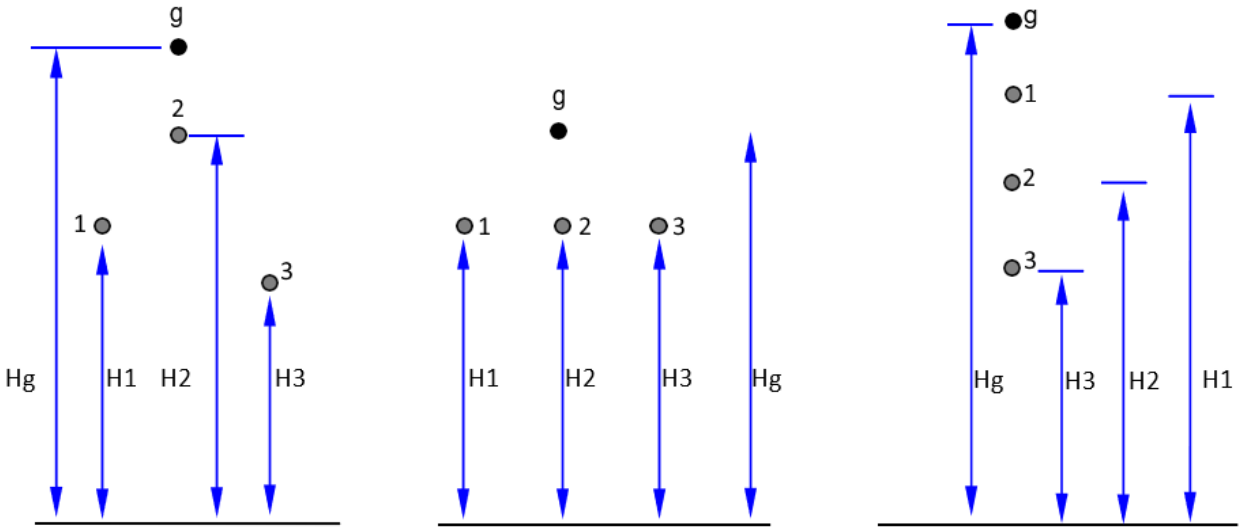


Figure B-12. Height of Phase Conductors and Ground Wire

DSLSUBT-08: Subtransmission Line Data – Underground/Submarine Cable

The data for the Underground Subtransmission Lines are the same as the requirements for the Primary Distribution Line Underground/Submarine Cable except for the Subtransmission Distribution Line Segment ID.

DSLSUBT-09: Substation Power Transformer – Two Winding

Substation Power Transformer ID
Specify the unique ID for the Substation Power Transformer using up to 25 alphanumeric characters.

From Primary Bus ID
Specify the Bus ID where the primary of the Substation Power Transformer is connected.

To Secondary Bus ID
Specify the Bus ID where the secondary of the Substation Power Transformer is connected.

Core Structure
Specify the Core Structure of the Substation Power Transformer:

1 – if not known
2 – for Shell Type Transformer
3 – for 3-legged Core Type Transformer
4 – for 4-legged Core Type Transformer
5 – for 5-legged Core Type Transformer

Method of Cooling
Specify the method of cooling of the Substation Power Transformer (e.g., OA, OA/FA, etc.)

KVA Rating (Primary)
Specify the Rated Capacity of the primary winding of the Substation Power Transformer in kVA.

KVA Rating (Secondary)
Specify the Rated Capacity of the secondary winding of the Substation Power Transformer in kVA.

Max KVA (Primary)
Specify the Maximum Capacity of the primary winding of the Substation Power Transformer in kVA where the transformer has forced cooling system.

Max KVA (Secondary)
Specify the Maximum Capacity of the secondary winding of the Substation Power Transformer in kVA where the transformer has forced cooling system.

KV Rating (Primary)
Specify the voltage rating of the primary winding of the Substation Power Transformer in kV.

KV Rating (Secondary)

Specify the voltage rating of the secondary winding of the Substation Power Transformer in kV.

Connection (Primary)
Specify the primary winding connection of the Substation Power Transformer:
DELTA or WYE

Connection (Secondary)
Specify the secondary winding connection of the Substation Power Transformer:
DELTA or WYE

Grounding (Primary)
Specify the grounding connection of the Substation Power Transformer at the primary side:
0 - Ungrounded
1 - Solidly Grounded
2 - Low Resistance Grounded
3 - High Resistance Grounded
4 - Reactance Grounded

Grounding (Secondary)
Specify the grounding connection of the Substation Power Transformer at the secondary side:
0 - Ungrounded
1 - Solidly Grounded
2 - Low Resistance Grounded
3 - High Resistance Grounded
4 - Reactance Grounded

Tap Changer Type
Specify the type of Tap Changer either:
Fixed - for Off-Load, Manual On-Load, and No Tap Changer
Automatic - for Automatic Load Tap Changer

Winding w/ Auto LTC
Specify the winding where automatic Load Tap Changing operation takes place:
PRI – for primary winding;
SEC – for secondary winding;
TER – for tertiary winding; or

NA – if not applicable.

Tap kV Setting (Primary)
Specify the Tap Voltage Setting in kV at the primary side. Enter the rated voltage if not applicable.

Tap kV Setting (Secondary)
Specify the Tap Voltage Setting in kV at the secondary side. Enter the rated voltage if not applicable.

Impedance (%Z)
Specify the Percent Impedance (%Z) of the transformer taken from the nameplate of the transformer. Use typical value if data is not available.

X/R Ratio
Specify the X/R ratio of the transformer impedance. Use typical value if data is not available.

No-Load Loss (kW)
Specify the No-load loss of the Substation Power Transformer in kW. Use typical value if data is not available.

Exciting Current (%)
Specify the Exciting Current of the Substation Power Transformer in percent (%) of the rated current. Use typical value if data is not available.

DSLSUBT-10: Substation Power Transformer – Three Winding

Substation Power Transformer ID
Specify the unique ID for the Substation Power Transformer using up to 25 alphanumeric characters.

From Primary Bus ID
Specify the Bus ID where the primary of the Substation Power Transformer is connected.

To Secondary Bus ID
Specify the Bus ID where the secondary of the Substation Power Transformer is connected.

To Tertiary Bus ID
Specify the Bus ID where the tertiary of the Substation Power Transformer is connected.

Core Structure
Specify the Core Structure of the Substation Power Transformer:

1 – if not known
2 – for Shell Type Transformer
3 – for 3-legged Core Type Transformer
4 – for 4-legged Core Type Transformer
5 – for 5-legged Core Type Transformer

Method of Cooling
Specify the method of cooling of the Substation Power Transformer (e.g., OA, OA/FA, etc.)

KVA Rating (Primary)
Specify the Rated Capacity of the primary winding of the Substation Power Transformer in kVA.

KVA Rating (Secondary)
Specify the Rated Capacity of the secondary winding of the Substation Power Transformer in kVA.

KVA Rating (Tertiary)
Specify the Rated Capacity of the tertiary winding of the Substation Power Transformer in kVA.

Max KVA (Primary)
Specify the Maximum Capacity of the primary winding of the Substation Power Transformer in kVA where the transformer has forced cooling system.

Max KVA (Secondary)
Specify the Maximum Capacity of the secondary winding of the Substation Power Transformer in kVA where the transformer has forced cooling system.

Max KVA (Tertiary)

Specify the Maximum Capacity of the tertiary winding of the Substation Power Transformer in kVA where the transformer has forced cooling system.

KV Rating (Primary)
Specify the voltage rating of the primary winding of the Substation Power Transformer in kV.

KV Rating (Secondary)
Specify the voltage rating of the secondary winding of the Substation Power Transformer in kV.

KV Rating (Tertiary)
Specify the voltage rating of the Tertiary winding of the Substation Power Transformer in kV.

Connection (Primary)
Specify the primary winding connection of the Substation Power Transformer:
DELTA or WYE

Connection (Secondary)
Specify the secondary winding connection of the Substation Power Transformer:
DELTA or WYE

Connection (Tertiary)
Specify the tertiary winding connection of the Substation Power Transformer:
DELTA or WYE

Grounding (Primary)
Specify the grounding connection of the Substation Power Transformer at the primary side:
0 - Ungrounded
1 - Solidly Grounded
2 - Low Resistance Grounded
3 - High Resistance Grounded
4 - Reactance Grounded

Grounding (Secondary)
Specify the grounding connection of the Substation Power Transformer at the secondary side:
0 - Ungrounded
1 - Solidly Grounded

2 - Low Resistance Grounded
3 - High Resistance Grounded
4 - Reactance Grounded

Grounding (Tertiary)
Specify the grounding connection of the Substation Power Transformer at the tertiary side:

0 - Ungrounded
1 - Solidly Grounded
2 - Low Resistance Grounded
3 - High Resistance Grounded
4 - Reactance Grounded

Tap Changer Type
Specify the type of Tap Changer either:

Fixed - for Off-Load, Manual On-Load, and No Tap Changer
Automatic - for Automatic Load Tap Changer

Winding w/ Auto LTC
Specify the winding where automatic Load Tap Changing operation takes place:

PRI – for primary winding;
SEC – for secondary winding;
TER – for tertiary winding; or
NA – if not applicable.

Tap kV Setting (Primary)
Specify the Tap Voltage Setting in kV at the primary side. Enter the rated voltage if not applicable.

Tap kV Setting (Secondary)
Specify the Tap Voltage Setting in kV at the secondary side. Enter the rated voltage if not applicable.

Tap kV Setting (Tertiary)
Specify the Tap Voltage Setting in kV at the tertiary side. Enter the rated voltage if not applicable.

DSLSUBT-12: Subtransmission Service Drop Data – Overhead

The Subtransmission Service Drop is the conductor extended from the Subtransmission Line to the Subtransmission load service entrance. The data for the Overhead Subtransmission Load Service Drops are the same as the requirements for the Overhead Primary Customer Service Drop ID except for the Subtransmission Load Service Drop ID.

DSLSUBT-13: Subtransmission Service Drop Data – Underground

The data for the Underground Subtransmission Service Drops are the same as the requirements for the Underground Primary Distribution Line except for the Subtransmission Load Service Drop ID.

DSLSUBT-13: Voltage Regulator Data

Voltage Regulator ID
Specify the unique ID for the Voltage Regulator using up to 25 alphanumeric characters.

Bus ID
Specify the Bus ID of the source side of the Voltage Regulator.

To Bus ID
Specify the Bus ID of the load side of the Voltage Regulator.
Regulated Bus ID
Specify the Bus ID of the regulating point (Bus or Node) whose voltage is being controlled by the Voltage Regulator.

Phase Type
Specify the type of Voltage Regulator:

1 – Single phase
2 – Two single phase
3 – Three-phase, gang operated
4 – Three single phase, independently operated

Phasing
Specify the Phasing of the Voltage Regulator:

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for Multi-grounded Three- Phase System;
b) ABN, BAN, BCN, CBN, CAN or ACN for Multi-grounded V-Phase System;
c) AN, BN or CN for Multi-grounded Single-Phase System;
d) ABC, ACB, BCA, BAC, CAB, CBA for Uni-grounded Three-Phase System;
e) AB, BA, BC, CB, CA or AC for Uni-grounded Single-Phase System;

Phase Sense
Specify the phase where the Voltage Sensor (PT) is installed:

A – if Phase A
B – if Phase B
C – if Phase C

kVA Rating
Specify the Rated Capacity of the Voltage Regulator in kVA.

kV Rating
Specify the voltage Rating of the Voltage Regulator in kV.

Voltage (120V base)
Specify the desired voltage (on 120-volt base) to be held by the Voltage Regulator at the regulating point (e.g., 124 volts).

Bandwidth (120V base)
Specify the voltage level tolerance of the Voltage Regulator on 120-volt base (e.g., 2.0 volts):

R-Setting Phase A
Specify the Compensator R-dial setting (i.e., the equivalent resistance to the regulating point calibrated in volts) of the Voltage Regulator in Phase A. Enter “0.0” if not applicable.

R-Setting Phase B
Specify the Compensator R-dial setting (i.e., the equivalent resistance to the regulating point calibrated in volts) of the Voltage Regulator in Phase B. Enter “0.0” if not applicable.

R-Setting Phase C

Specify the Compensator R-dial setting (i.e., the equivalent resistance to the regulating point calibrated in volts) of the Voltage Regulator in Phase C. Enter “0.0” if not applicable.

X-Setting Phase A
Specify the Compensator X-dial setting (i.e., the equivalent reactance to the regulating point calibrated in volts) of the Voltage Regulator in Phase A. Enter “0.0” if not applicable.

X-Setting Phase B
Specify the Compensator X-dial setting (i.e., the equivalent reactance to the regulating point calibrated in volts) of the Voltage Regulator in Phase B. Enter “0.0” if not applicable.

X-Setting Phase C
Specify the Compensator X-dial setting (i.e., the equivalent reactance to the regulating point calibrated in volts) of the Voltage Regulator in Phase C. Enter “0.0” if not applicable.

Primary Current Rating (A)
Specify the primary current rating of the Current Transformer used for the Voltage Regulator. The CT secondary current is assumed 1 Ampere.

PT Ratio
Specify the voltage ratio of the Potential Transformer used for the Voltage Regulator. Usually the PT secondary voltage of Voltage Regulator is 120 volts. For example, a PT rated 13,200/120 volts has a PT Ratio of 110.

No-Load Loss (kW)
Specify the No-Load (fixed) loss of the Voltage Regulator in kW.

Exciting Current (%)
Specify the exciting current of the Voltage Regulator in percent (%) of the rated current.

DSLSUBT-14: Shunt Capacitor Data

Shunt Capacitor ID
Specify the unique ID for the Shunt Capacitor using up to 25 alphanumeric characters.

Bus Connected (Bus ID)
Specify the Bus ID of the Bus or Node where the Shunt Capacitor is connected.

Phase Type
Specify the construction type of Shunt Capacitor:

1 – Single-phase Shunt Capacitor
2 – Two (2) single-phase Shunt Capacitors
3 – Three-phase Shunt Capacitor
4 – Three (3) single-phase Shunt Capacitors

Phasing
Specify the Phasing of the Shunt Capacitor:

a) ABC for Three-Phase
b) AB, BC or CA for V-Phase
c) A, B, C or Single-Phase

Voltage Rating (kV)
Specify the Voltage Rating of Shunt Capacitor in kV.

KVAR Rating Phase A
Specify reactive Power Rating in kVArs of the Shunt Capacitor in phase A.

KVAR Rating Phase B
Specify reactive Power Rating in kVArs of the Shunt Capacitor in phase B.

KVAR Rating Phase C
Specify reactive Power Rating in kVArs of the Shunt Capacitor in phase C.

Power Loss (Watts)
Specify the Power Losses of the Shunt Capacitor per phase in Watts. Use Typical Value if Power Loss data of the Shunt Capacitor is not known.

DSLSUBT-15: Shunt Inductor Data

Shunt Inductor ID
Specify the unique ID for the Shunt Inductor using up to 25 alphanumeric characters.

Bus Connected (Bus ID)
Specify the Bus ID of the Bus or Node where the Shunt Inductor is connected.
Phase Type
Specify the construction type of Shunt Inductor:

1 – Single-phase Shunt Inductor
2 – Two (2) single-phase Shunt Inductors
3 – Three-phase Shunt Inductor
4 – Three (3) single-phase Shunt Inductors

Phasing
Specify the Phasing of the Shunt Inductors:

a) ABC for Three-Phase
b) AB, BC or CA for V-Phase
c) A, B, C or Single-Phase

Voltage Rating (kV)
Specify the Voltage Rating of the Shunt Inductor in kV.

Resistance Phase A (Ohms)
Specify the Resistance in Ohms of the Shunt Inductor in Phase A. Use typical values if the resistance of the Shunt Inductor is not available.

Resistance Phase B (Ohms)
Specify the Resistance in Ohms of the Shunt Inductor in Phase B. Use typical values if the resistance of the Shunt Inductor is not available.

Resistance Phase C (Ohms)
Specify the Resistance in Ohms of the Shunt Inductor in Phase C. Use typical values if the resistance of the Shunt Inductor is not available.

Reactance Phase A (Ohms)
Specify the Reactance in Ohms of the Shunt Inductor in Phase A.

Reactance Phase B (Ohms)
Specify the Reactance in Ohms of the Shunt Inductor in Phase B.

Reactance Phase C (Ohms)
Specify the Reactance in Ohms of the Shunt Inductor in Phase C.

DSLSUBT-16: Series Inductor Data

Series Inductor ID
Specify the unique ID for the Series Inductor using up to 25 alphanumeric characters.

From Bus ID
Specify the Bus ID of the source side of the Series Inductor.
From Bus ID
Specify the Bus ID of the load side of the Series Inductor.

Phase Type
Specify the construction type of Series Inductor:

1 – One (1) Single-phase Series Inductor
2 – Two (2) single-phase Series Inductors
3 – One (1) Three-phase Series Inductor
4 – Three (3) single-phase Series Inductors

Phasing
Specify the Phasing of the Series Inductors:

a) ABCN, ACBN, BCAN, BACN, CABN or CBAN for Multi-grounded Three- Phase System;
b) ABN, BAN, BCN, CBN, CAN or ACN for Multi-grounded V-Phase System;
c) AN, BN or CN for Multi-grounded Single-Phase System;
d) ABC, ACB, BCA, BAC, CAB, CBA for Uni-grounded Three-Phase System;
e) AB, BA, BC, CB, CA or AC for Uni-grounded Single-Phase System;

Voltage Rating (kV)
Specify the Voltage Rating of the Series Inductor in kV.

Resistance Phase A (Ohms)
Specify the Resistance in Ohms of the Series Inductor in Phase A. Use typical values if the resistance of the Series Inductor is not available.

Resistance Phase B (Ohms)

Specify the Resistance in Ohms of the Series Inductor in Phase B. Use typical values if the resistance of the Series Inductor is not available.

Resistance Phase C (Ohms)
Specify the Resistance in Ohms of the Series Inductor in Phase C. Use typical values if the resistance of the Series Inductor is not available.

Reactance Phase A (Ohms)
Specify the Reactance in Ohms of the Series Inductor in Phase A.

Reactance Phase B (Ohms)
Specify the Reactance in Ohms of the Series Inductor in Phase B.

Reactance Phase C (Ohms)
Specify the Reactance in Ohms of the Series Inductor in Phase C.

DSLSUBT-17: Generator Data

Generator ID
Specify the unique ID of the generator in the distribution. Use up to 25 alphanumeric characters.

Bus Connected (Bus ID)
Specify the Bus ID where the distributed generator is connected.

KVA Rating
Specify the rated capacity of the distributed generator in kVA.

KW Rating
Specify the rated real power of the generator in kW.

Minimum KVAR
Specify the minimum reactive power limit that the generator can inject in kVArs. Specify a negative sign if the generator can absorb reactive power.

Maximum KVAR
Specify the maximum reactive power limit that the generator can inject in kVArs.

Voltage Rating (kV)
Specify the voltage rating of the primary winding of the distributed generator in kV.
The voltage rating should be taken from the nameplate and not the resulting line-to-line voltage of the generator.

Dispatch Curve ID
Specify the Dispatch Curve ID that will be applied to the generator. All Dispatch Curve ID must be defined in the Dispatch Curve Data. Use up to 25 alphanumeric characters.

DG Model
Specify the distributed generator model used in the simulation. Both models have constant real power (P) that will follow the corresponding dispatch curve. The defined models are:

PQ – Constant real and reactive power (P, Q). Reactive Power values follow its corresponding Reactive Power Curve

PV – Constant real power and voltage (P, V). Source voltage values follow its corresponding Voltage Profile

Voltage Profile ID
Specify the Voltage Profile ID that will be applied to the generator for DGs modeled as PV. All Voltage Profile ID must be defined in the Voltage Profile Data. Put “NA” if the DG is modeled as PQ. Use up to 25 alphanumeric characters.

Reactive Power Curve ID
Specify the Reactive Power Curve ID that will be applied to the generator for DGs modeled as PQ. All Reactive Power Curve ID must be defined in the Reactive Power Curve Data. Put “NA” if the DG is modeled as PV. Use up to 25 alphanumeric characters.

Phasing
Specify the Phasing of the Generator:

ABC, ACB, BCA, BAC, CAB, or CBA.

Connection
Specify the connection of the Generator:

1 – Wye Ground
2 – Wye
3 – Delta

Positive Sequence Resistance (Ohms)
Specify the Positive Sequence Resistance in Ohms of the Generator. Use a typical value if the resistance of the Generator is not available.

Positive Sequence Reactance (Ohms)
Specify the Positive Sequence Reactance in Ohms of the Generator. Use a typical value if the reactance of the Generator is not available.

Negative Sequence Resistance (Ohms)
Specify the Negative Sequence Resistance in Ohms of the Generator. Use a typical value if the resistance of the Generator is not available.

Negative Sequence Reactance (Ohms)
Specify the Negative Sequence Reactance in Ohms of the Generator. Use a typical value if the reactance of the Generator is not available.

Zero Sequence Resistance (Ohms)
Specify the Zero Sequence Resistance in Ohms of the Generator. Use a typical value if the resistance of the Generator is not available.

Zero Sequence Reactance (Ohms)
Specify the Zero Sequence Reactance in Ohms of the Generator. Use a typical value if the reactance of the Generator is not available.

DSLSUBT-18: Generator Dispatch Curve

Dispatch Curve ID
Specify the unique ID of the generator dispatch curve in the distribution. This must be corresponding to the Dispatch Curve ID specified in the Generator Data. Use up to 25 alphanumeric characters.

Generator Type
Specify the generator type for regulation purposes. (e.g., Solar, Wind, etc.).

Description

Describe the Dispatch Curve. For example, Day-time is a dispatch schedule for the Solar PV generator types.

Hour 1 to Hour 24
Specify the normalized rated real power percentage from Hour 1 to Hour 24 of the Dispatch Curve in decimals.

DSLSUBT-19: Voltage Profile

Voltage Profile ID
Specify the ID of the voltage profile of the root source and generators modeled as PV in the distribution. This must be corresponding to the Voltage Profile ID specified in the Generator Data. The first entry should be for the 24-hour Source Voltage. Use up to 25 alphanumeric characters.

Description
Describe the Voltage Profile. For example, Source_Voltage is a constant voltage profile of 1.0 per unit.

Hour 1 to Hour 24
Specify the normalized hourly voltage values from Hour 1 to Hour 24 of the Voltage Profile in per unit.

DSLSUBT-20: Generator Reactive Power Curve

Reactive Power Curve ID
Specify the ID of the reactive power curve of the generator modeled as PQ in the distribution. This must be corresponding to the Reactive Power Curve ID specified in the Generator Data. Use up to 25 alphanumeric characters.

Description
Describe the Reactive Power Curve. For example, Unity_pf is a constant value of 0 kVAr for a unity power factor generator.

Hour 1 to Hour 24
Specify the hourly reactive power injection of the generator from Hour 1 to Hour 24 of the Reactive Power Curve in kVars.

Reliability Sheets for Feeder

The Reliability Data Requirements templates are the additional sheets to the DSL Template to accommodate data for reliability simulations using the Reliability Analysis Tool:

a) RID-21: Primary Distribution Sectionalizing or Switching Device Data
b) RID-22: Primary Distribution Predictive Reliability Data
c) RID-23: Primary Distribution Interruption Event Data

DSL-RID-21: Primary Distribution Sectionalizing or Switching Device Data

Primary Distribution Device ID
Specify the unique ID of the sectionalizing or switching device in the primary distribution. Use up to 25 alphanumeric characters.

Primary Distribution Line Segment ID
Specify the ID of the primary distribution line segment, from the relevant worksheet (either from “Primary Distribution Line Data – Overhead” or from “Primary Distribution Line Data – Underground”).

Device Type Code
Specify the device code corresponding to the device installed at this location. Use the following device codes:

201 – Circuit breaker (or recloser with reclosing capability deactivated)
202 – Recloser (or circuit breaker with reclosing capability activated)
203 – Fuse-blow fuse
204 – Sectionalizer
205 – Switch (normally-closed)
206 – Network protectors
207 – Fuse-save fuse

Switching Time
If the device at this location is a switch, enter the switching time over which this switch is expected to operate to restore service for permanent outages in hours (e.g., “0.50” if the expected switching time is 30 minutes). Otherwise, input “–1” (negative one).

DSL-RID-22: Primary Distribution Predictive Reliability Data

Primary Distribution Line Segment ID
Specify the ID of the Primary Distribution line segment, from the relevant worksheet (either from “Primary Distribution Line Data – Overhead” or from “Primary Distribution Line Data – Underground”).

Permanent Failure Rate (failures/year)
Input the expected amount of permanent failures per year for this primary distribution line segment only, excluding the failures for downstream segments.

Temporary Failure Rate (failures per year)
Input the expected amount of temporary failures per year for this primary distribution line segment only, excluding the failures for downstream segments.

Repair Time
Input the expected amount of time in hours for service restoration in the case when this segment suffers a permanent outage and repair will be needed to restore service. Use the expected value only for this segment and exclude the repair times for any downstream segment.

DSL-RID-23: Primary Distribution Interruption Event Data

Each event (e.g., the opening of an interrupting device, isolation of a faulted segment, step restoration, unrelated interruptions) must be included as one data entry. The status of the devices (interrupting device or terminal point) must be true throughout the event (start to restoration).

Event ID
Specify the unique ID for the Interruption Event using up to 25 alphanumeric characters. Reserved numbers are from 0 to 500, including any those with leading zeroes.

Example: 00 and 0500 are banned, but A00 and A0500 are allowed.

Affected Town(s) or City(ies)
Specify the town(s) or city(ies) affected by the interruption event.

Affected Barangay(s) or Subdivision(s)
Specify the barangay(s) or subdivision(s) affected by the interruption event.

Ckt. No.
Specify the circuit number affected by the interruption event.

Start Date (mm/dd/yy)
Specify the date during which the event started, using the mm/dd/yy format. Ensure that the spreadsheet program uses the 01 January 1900 as the reference date.

Start Time (hh:mm) 24-hour format
Specify the time during which the event started, using the HH:MM 24-hour format.

Restoration Date (mm/dd/yy)
Specify the date during which service was restored, using the mm/dd/yy format. Ensure that the spreadsheet program uses 01 January 1900 as the reference date.

Restoration Time (hh:mm) 24-hour format
Specify the time during which service was restored, using the HH:MM 24-hour format.

Duration (minutes)
Specify the duration of the outage in minutes.

Code for Cause of Interruption
Indicate the code corresponding to the cause of the interruption. Use the following cause codes:

1 – Human being
2 – Lightning
3 – Major storm disaster
4 – Scheduled interruption
5 – Trees
6 – Overload
7 – Error
8 – Supply
9 – Equipment
10 – Others
11 – Unknown
12 – Earthquakes

Code for Weather during Interruption
Indicate the code corresponding to the weather during the interruption event. Use the following interruption weather codes:

101 – Wind
102 – Lightning
103 – Rain
104 – Clear day
105 – Typhoon

Device ID of Interrupting Device
Indicate the ID of the device that responded (opened). Customers who are located downstream of the interrupting device are interrupted. This should be correlated with device IDs contained in the “PriDist – Sect and Sw Dev” worksheet.

Code for Equipment Failed
Indicate the code corresponding to the main equipment that failed and caused the interruption event. Use the following equipment codes:

302 – Primary line
303 – Secondary line
304 – Distribution Transformer
305 – Substation Power Transformer
306 – Splice
307 – Lightning arrester
308 – Switches
309 – Cross arm
310 – Pole
311 – Insulator
312 – Connector
313 – Others
314 – Unknown

Phase Affected
Specify the phases affected by the interrupting device:

A, B, C, AB, AC, BC, or ABC.

Device ID of Terminal Point
Indicate the ID of the device that acts as the terminal point of interruption. Customers who are located downstream of the terminal point are re-energized. This should be correlated with device IDs contained in the “PriDist – Sect and Sw Dev” worksheet. Otherwise, place a “0” (zero).

Create a new row if the start time (and date) and restoration time (and date) of the interrupting device are not the same as the start time (and date) and restoration time (and date) of the terminal point. You may add more rows for additional terminal points.

Phase of Terminal Point
Indicate the phase of the terminal point. Otherwise, place a “0” (zero).

Reliability Sheets for Subtransmission

The following are the additional sheets required for the Subtransmission Reliability Analysis Tool:

a) RID-21: Subtransmission Sectionalizing or Switching Device Data
b) RID-22: Subtransmission Predictive Reliability Data
c) RID-23: Subtransmission Interruption Event Data

DSLSUBT-RID-21: Subtransmission Sectionalizing or Switching Device Data

Subtransmission Device ID
Specify the unique ID of the sectionalizing or switching device in the subtransmission. Use up to 25 alphanumeric characters.

Subtransmission Line Segment ID
Specify the ID of the subtransmission line segment, from the relevant worksheet (either from "Subtransmission Line Data – Overhead" or from "Subtransmission Line Data – Underground").

Device Type Code
Specify the device code corresponding to the device installed at this location. Use the following device codes:

201 – Circuit breaker (or recloser with reclosing capability deactivated)
202 – Recloser (or circuit breaker with reclosing capability activated)
203 – Fuse-blow fuse
204 – Sectionalizer
205 – Switch (normally-closed)
206 – Network protectors
207 – Fuse-save fuse

Switching Time
If the device at this location is a switch, enter the switching time over which this switch is expected to operate to restore service for permanent outages in hours (e.g., "0.50" if the expected switching time is 30 minutes). Otherwise, input "–1" (negative one).

DSLSUBT-RID-22: Subtransmission Predictive Reliability Data

Subtransmission Line Segment ID
Specify the ID of the subtransmission line segment, from the relevant worksheet (either from "Subtransmission Line Data – Overhead" or from "Subtransmission Line Data – Underground").

Permanent Failure Rate (failures/year)
Input the expected amount of permanent failures per year for this subtransmission line segment only, excluding the failures for downstream segments.

Temporary Failure Rate (failures per year)
Input the expected amount of temporary failures per year for this subtransmission line segment only, excluding the failures for downstream segments.

Repair Time
Input the expected amount of time in hours for service restoration in the case when this segment suffers a permanent outage and repair will be needed to restore service. Use the expected value only for this segment and exclude the repair times for any downstream segment.

DSLSUBT-RID-23: Subtransmission Interruption Event Data

Each event (e.g., the opening of an interrupting device, isolation of a faulted segment, step restoration, unrelated interruptions) must be included as one data entry. The status of the devices (interrupting device or terminal point) must be true throughout the event (start to restoration).

Event ID
Specify the unique ID for the Interruption Event using up to 25 alphanumeric characters. Reserved numbers are from 0 to 500, including any those with leading zeroes.

Example: 00 and 0500 are banned, but A00 and A0500 are allowed.

Affected Town(s) or City(ies)
Specify the town(s) or city(ies) affected by the interruption event.

Affected Barangay(s) or Subdivision(s)
Specify the barangay(s) or subdivision(s) affected by the interruption event.

Ckt. No.
Specify the circuit number affected by the interruption event.

Start Date (mm/dd/yy)
Specify the date during which the event started, using the mm/dd/yy format. Ensure that the spreadsheet program uses the 01 January 1900 as the reference date.

Start Time (hh:mm) 24-hour format
Specify the time during which the event started, using the HH:MM 24-hour format.

Restoration Date (mm/dd/yy)
Specify the date during which service was restored, using the mm/dd/yy format. Ensure that the spreadsheet program uses the 01 January 1900 as the reference date.

Restoration Time (hh:mm) 24-hour format
Specify the time during which service was restored, using the HH:MM 24-hour format.

Duration (minutes)
Specify the duration of the outage in minutes.

Code for Cause of Interruption
Indicate the code corresponding to the cause of the interruption. Use the following cause codes:

1 – Human being
2 – Lightning
3 – Major storm disaster
4 – Scheduled interruption
5 – Trees
6 – Overload
7 – Error
8 – Supply
9 – Equipment
10 – Others
11 – Unknown
12 – Earthquakes

Code for Weather during Interruption
Indicate the code corresponding to the weather during the interruption event. Use the following interruption weather codes:

101 – Wind
102 – Lightning
103 – Rain
104 – Clear day
105 – Typhoon

Device ID of Interrupting Device
Indicate the ID of the device that responded (opened). Customers who are located downstream of the interrupting device are interrupted. This should be correlated with device IDs contained in the “Subtrans – Sect and Sw Dev” worksheet.

Code for Equipment Failed
Indicate the code corresponding to the main equipment that failed and caused the interruption event. Use the following equipment codes:

302/303 – Subtransmission line
304/305 – Substation Power Transformer
306 – Splice
307 – Lightning arrester
308 – Switches
309 – Cross arm
310 – Pole
311 – Insulator
312 – Connector
313 – Others
314 – Unknown

Phase Affected
Specify the phases affected by the interrupting device:

A, B, C, AB, AC, BC, or ABC.

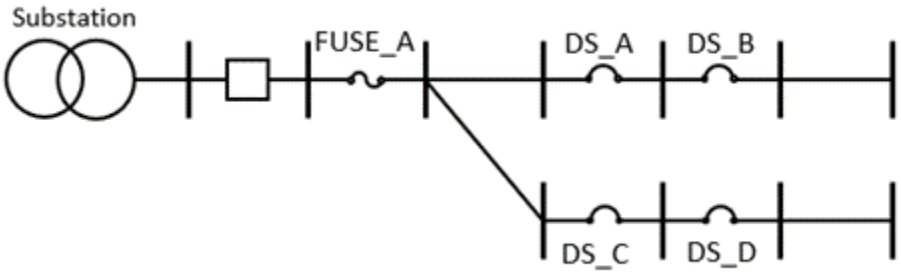
Device ID of 1st Terminal Point
Indicate the ID of the device that acts as the terminal point of interruption. Customers who are located downstream of the terminal point are re-energized. This should be correlated with device IDs contained in the “Subtrans – Sect and Sw Dev” worksheet. Otherwise, place a “0” (zero).

Create a new row if the start time (and date) and restoration time (and date) of the interrupting device are not the same as the start time (and date) and restoration time (and date) of the terminal point. You may add more rows for additional terminal points.

Phase of Terminal Point
Indicate the phase of the 1st terminal point. Otherwise, place a “0” (zero).

Interruption Events Sheet and Examples

Consider the network below:



If an opening or tripping of a sectionalizing or switching device interrupted its downstream customers, the event should be entered in the Interruption Events sheet. The interrupting device should be the starting point of power loss. Enter the time when it was opened as the Start Time (and Date) and the time when it was closed as the Restoration Time (and Date).

Example 1

20 October 2019	
Time Stamp	**Description**
1:00 PM	FUSE_A blew off.
1:35 PM	FUSE_A was replaced and re-energized.

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	13:00	10/20/19	13:35	…	FUSE_A	ABC	0	0

Example 2

20 October 2019

Time Stamp	Description
7:45 PM	DS_A was opened
8:05 PM	FUSE_A blew off
8:15 PM	DS_A was closed
8:30 PM	FUSE_A was replaced and re-energized

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	19:45	10/20/19	20:15	…	DS_A	ABC	0	0
…	10/20/19	20:05	10/20/19	20:30	…	FUSE_A	ABC	0	0

Example 3

20 October 2019	
Time Stamp	**Description**
10:10 AM	DS_A had one phase out (Phase A)
10:13 AM	FUSE_A blew off
11:00 AM	FUSE_A was replaced and re-energized
12:15 PM	One phase-out of DS_A was re-energized

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	10:10	10/20/19	12:15	…	DS_A	A	0	0

…	10/20/19	10:13	10/20/19	11:00	…	FUSE_A	ABC	0	0

If another device was opened to re-energize its downstream customers, enter the device ID as a Terminal Point. If the Terminal Point was opened at the Start Time (and Date) of the Interruption and the Terminal Point was closed on or after the Restoration Time (and Date) of the Interruption, enter the terminal point in the same row as the interruption.

Example 4

20 October 2019	
Time Stamp	**Description**
2:25 PM	DS_A was opened and the segment from DS_B to End was shifted to another feeder (opened DS_B)
3:15 PM	Segment from DS_A to End was normalized (closed DS_A, closed DS_B)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	14:25	10/20/19	15:15	…	DS_A	ABC	DS_B	ABC

If the Terminal Point was opened after the Start Time of the Interruption, create a new row. Copy all the fields from the previous row. Replace the Event ID with a new one. Edit the Terminal Point ID and Phase. Replace the Start Time (and Date) with the time when the Terminal Point was opened. If the Terminal Point was closed on or after the Restoration Time (and Date) of the Interruption, do not change the Restoration Time (and Date). Otherwise, replace the Restoration Time (and Date) with the time when the Terminal Point was closed.

Example 5

20 October 2019	
Time Stamp	**Description**
4:15 PM	FUSE_A blew off

4:35 PM	Segment from DS_A to End was shifted to another feeder (opened DS_A)
5:10 PM	FUSE_A was replaced and re-energized and the segment from DS_A to End was normalized (closed DS_A)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	16:15	10/20/19	17:10	…	FUSE_A	ABC	0	0
…	10/20/19	16:35	10/20/19	17:10	…	FUSE_A	ABC	DS_A	ABC

Example 6

20 October 2019	
Time Stamp	**Description**
9:20 AM	FUSE_A blew off
9:35 AM	Segment from DS_A to End was shifted to another feeder
10:00 AM	FUSE_A was replaced and re-energized
10:05 AM	Segment from DS_A to End was normalized (closed DS_A)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	09:20	10/20/19	10:00	…	FUSE_A	ABC	0	0
…	10/20/19	09:35	10/20/19	10:00	…	FUSE_A	ABC	DS_A	ABC

You may add more rows for additional terminal points.

Example 7

20 October 2019	
Time Stamp	**Description**
8:05 AM	FUSE_A blew off
8:25 AM	Segment from DS_A to End was shifted to another feeder (opened DS_A)
8:27 AM	Segment from DS_C to End was shifted to another feeder (opened DS_C)
8:45 AM	FUSE_A was replaced and re-energized
9:00 AM	Segment from DS_A to End was normalized (closed DS_A)
9:03 AM	Segment from DS_C to End was normalized (closed DS_C)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	08:05	10/20/19	08:45	…	FUSE_A	ABC	0	0
…	10/20/19	08:25	10/20/19	08:45	…	FUSE_A	ABC	DS_A	ABC
…	10/20/19	08:27	10/20/19	08:45	…	FUSE_A	ABC	DS_C	ABC

If another device was open when a downstream Terminal Point was closed, the device should be entered as the new Terminal Point. Copy all the fields from the row of the previous Terminal Point. Replace the Event ID with a new one. Edit the Terminal Point ID and Phase. Replace the Start Time (and Date) with the Restoration Time (and Date) of the previous row. If the new Terminal Point was closed on or after the Restoration Time (and Date) of the Interruption, do not change the Restoration Time (and Date). Otherwise, replace the Restoration Time (and Date) with the time when the new Terminal Point was closed.

Example 8

20 October 2019	
Time Stamp	**Description**
11:55 AM	FUSE_A blew off
12:10 PM	Segment from DS_B to End was shifted to another feeder (opened DS_B)
12:20 PM	DS_A was opened to isolate the faulted segment
12:27 PM	DS_B was closed
1:05 PM	FUSE_A was replaced and re-energized
1:10 PM	Segment from DS_A to End was normalized (closed DS_A)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	11:55	10/20/19	13:05	…	FUSE_A	ABC	0	0
…	10/20/19	12:10	10/20/19	12:27	…	FUSE_A	ABC	DS_B	ABC
…	10/20/19	12:27	10/20/19	13:05	…	FUSE_A	ABC	DS_A	ABC

Example 9

20 October 2019	
Time Stamp	**Description**
7:30 AM	FUSE_A blew off
7:55 AM	Segment from DS_B to End was shifted to another feeder (opened DS_B)

7:59 AM	Segment from DS_D to End was shifted to another feeder (opened DS_D)
8:20 AM	DS_A and DS_C were opened to isolate the faulted segment
8:23 AM	DS_B was closed
8:26 AM	DS_D was closed
8:50 AM	FUSE_A was replaced and re-energized
8:55 AM	Segment from DS_A to End was normalized (closed DS_A)
5:58 AM	Segment from DS_C to End was normalized (closed DS_C)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	07:30	10/20/19	08:50	…	FUSE_A	ABC	0	0
…	10/20/19	07:55	10/20/19	08:23	…	FUSE_A	ABC	DS_B	ABC
…	10/20/19	07:59	10/20/19	08:26	…	FUSE_A	ABC	DS_D	ABC
…	10/20/19	08:23	10/20/19	08:50	…	FUSE_A	ABC	DS_A	ABC
…	10/20/19	08:26	10/20/19	08:50	…	FUSE_A	ABC	DS_C	ABC

If the interruption device was closed but a downstream device (that was not a terminal point) was open, the downstream customers of the open device are still unenergized. The open device should be the new interrupting device. Copy all the fields from the previous row. Replace the Event ID with a new one. Edit the Interrupting Device ID and Phase. Replace the Start Time (and Date) with the Restoration Time (and Date) of the previous row. Replace the Restoration Time (and Date) with the time when the new Interrupting Device was closed.
If a terminal point was still open when the Interrupting Device was replaced, apply the instructions discussed in the previous examples.

Example 10

20 October 2019	
Time Stamp	**Description**
9:18 PM	FUSE_A blew off
9:25 PM	DS_A and DS_B are opened to isolate the faulted segment
9:30 PM	Segment from DS_B to End was shifted to another feeder
9:43 PM	FUSE_A was replaced and re-energized
10:10 PM	DS_A was closed
10:12 PM	Segment from DS_B to End was normalized (closed DS_B)

…	Start Date	Start Time	Restoration Date	Restoration Time	…	Interrupting Device ID	Phase/s Affected	Terminal Point ID	Phase/s of Terminal Point
…	10/20/19	21:18	10/20/19	21:43	…	FUSE_A	ABC	0	0
…	10/20/19	21:30	10/20/19	21:43	…	FUSE_A	ABC	DS_B	ABC
…	10/20/19	21:43	10/20/19	22:10	…	DS_A	ABC	DS_B	ABC

Always check if the devices entered in both the Interrupting Device ID and Terminal Point ID columns were open between the time

Simulation Parameters Sheet for Short Circuit

This data describes the parameters that will be used in the simulation of short-circuit calculation. The system before the connection point of the feeder is considered as the source.

Positive Sequence Source Resistance (Ohms)
Specify the Positive Sequence Resistance in Ohms of the Source.

Positive Sequence Source Reactance (Ohms)
Specify the Positive Sequence Reactance in Ohms of the Source.

Negative Sequence Source Resistance (Ohms)
Specify the Negative Sequence Resistance in Ohms of the Source.

Negative Sequence Source Reactance (Ohms)
Specify the Negative Sequence Reactance in Ohms of the Source.

Zero Sequence Source Resistance (Ohms)
Specify the Zero Sequence Resistance in Ohms of the Source.

Zero Sequence Source Reactance (Ohms)
Specify the Zero Sequence Reactance in Ohms of the Source.

Fault Resistance (Ohms)
Specify the Resistance in Ohms of the Fault.

Fault Reactance (Ohms)
Specify the Reactance in Ohms of the Fault.

------- *NOTHING FOLLOWS* -------